M3. Generative Models with Other Statistical Distance

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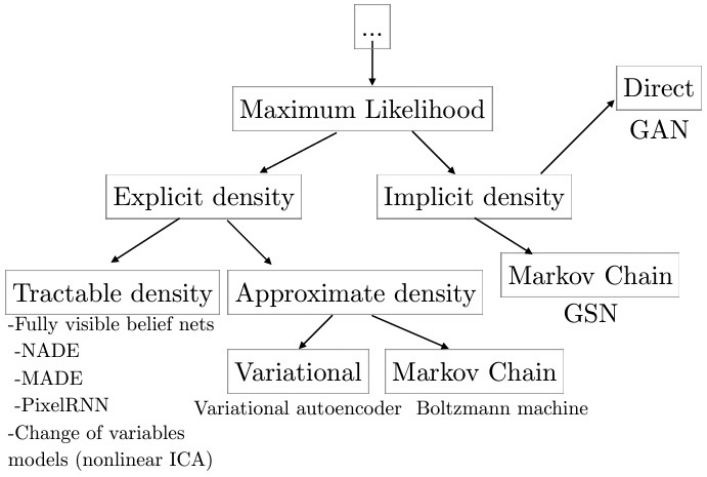
Department of Mathematical Sciences

FS26 MTH345: Deep Generative Models: A Statistical Perspective



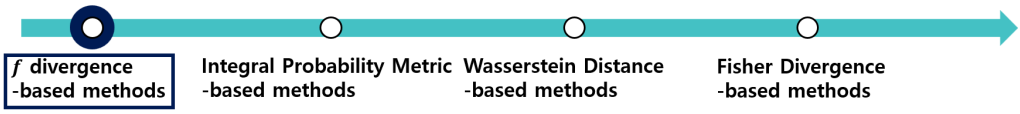


Introduction



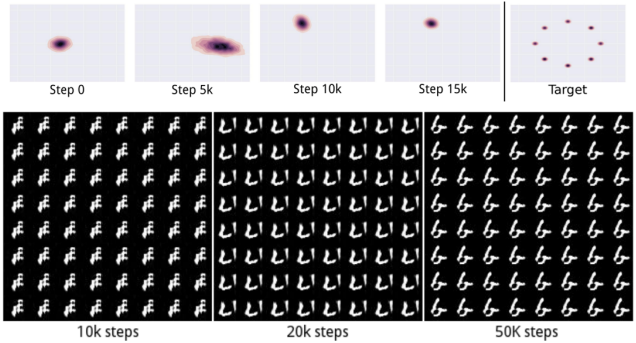
The figure is from Goodfellow (2016).

Introduction





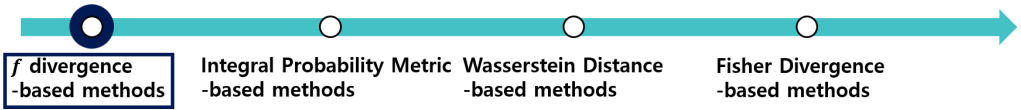
(a) Blurry generation results from VAEs



(b) Mode collapse results from GANs

Images are edited from Tolstikhin et al. (2018) and Metz et al. (2017).

Introduction



- The mode collapse phenomenon in GANs demonstrates that p_θ fails to capture the support of p.
- Several works have criticized the f-divergence,

$$\mathcal{D}_f(p \parallel p_\theta) = \int_{\mathbb{R}^d} f\left(\frac{p(\boldsymbol{x})}{p_\theta(\boldsymbol{x})}\right) p_\theta(\boldsymbol{x})\, d\boldsymbol{x}, \tag{1}$$

pointing out that the density ratio, $p(\boldsymbol{x})/p_\theta(\boldsymbol{x})$, is not well-defined when p places density (or mass) on a region where p_θ has (almost) zero.

Lecture Topics

- In this module, we study several alternative statistical distances in the deep generative model literature.
- Recommended reference materials include:
  - Chap. 9: Gretton et al. (2012), "A kernel two-sample test," *The Journal of Machine Learning Research*.
  - Chap. 10: Santambrogio (2015), "Optimal Transport for Applied Mathematicians: Calculus of Variations, PDEs, and Modeling," *Birkhäuser*.
  - Chap. 11: Hyvärinen (2005), "Estimation of non-normalized statistical models by score matching," *The Journal of Machine Learning Research*.

Lecture Topics

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Likelihood Maximization Principle
4. Basics of Probabilistic Modelling

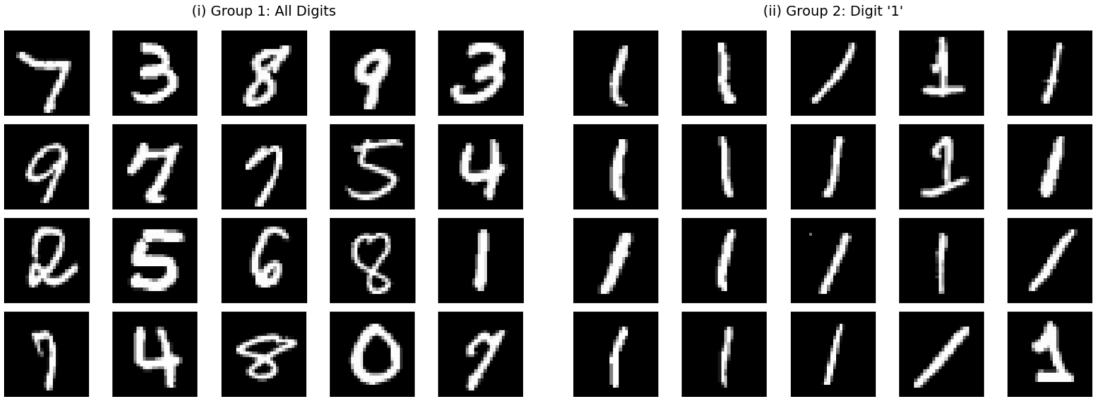
M2. Likelihood Ratio-based Generative Models

5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

M3. Generative Models with Other Statistical Distance

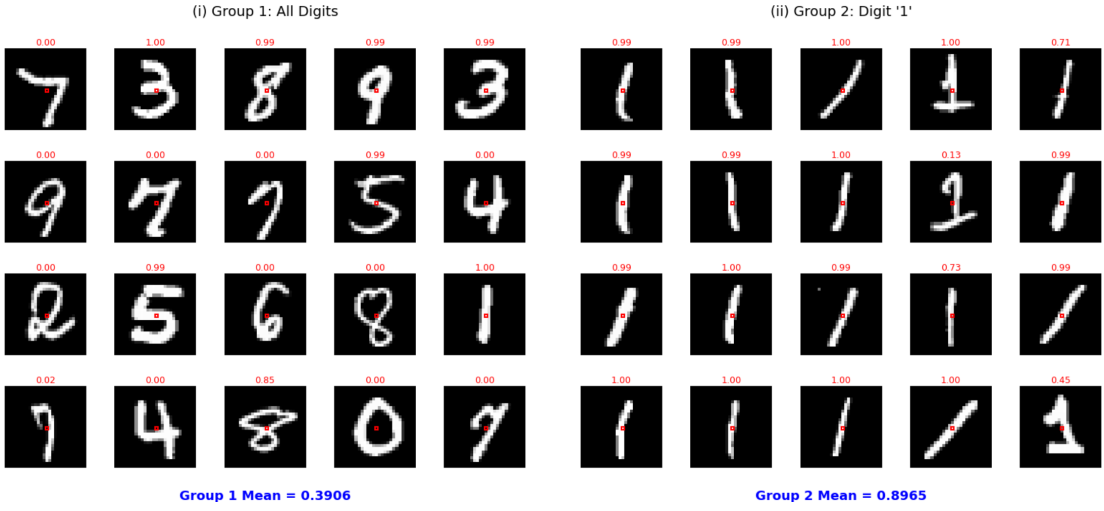
9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Motivation

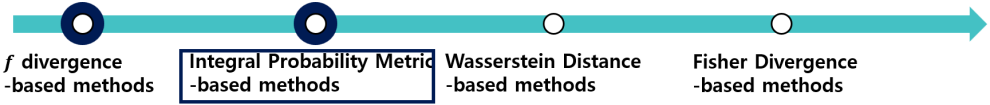


- **Q**: How to quantify the difference between two data collections?

Motivation


(i) Group 1: All Digits
(ii) Group 2: Digit '1'
Group 1 Mean = 0.3906
Group 2 Mean = 0.8965

Motivation

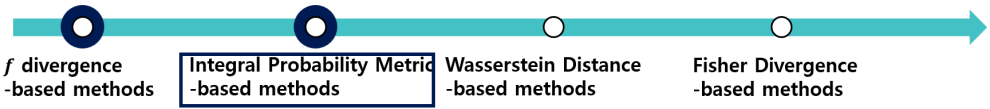


- As an alternative to density ratios, a line of work has proposed focusing on discrepancy measures that are effective regardless of the differences between the supports of p and p_θ.
- For example, Generative Moment Matching Networks (Li et al., 2015) aim to minimize:

$$\left\| \int_{\mathbb{R}^d} \varphi(\boldsymbol{x}) p(\boldsymbol{x})\, d\boldsymbol{x} - \int_{\mathbb{R}^d} \varphi(\boldsymbol{x}) p_\theta(\boldsymbol{x})\, d\boldsymbol{x} \right\|^2, \tag{2}$$

  where the integrated terms $\varphi(\boldsymbol{x})$ represent vectors of finite moments.
- This loss function is a special case of integral probability metrics, $\gamma_{\mathcal{F}}(p, p_\theta)$, where $\mathcal{F}$ denotes a set of summary statistics functions, such as moments.

Integral Probability Metric



- The Integral Probability Metric (IPM) between $p(\boldsymbol{x})$ and $q(\boldsymbol{x})$ is defined as:

$$\gamma_{\mathcal{F}}(p, q) := \sup_{f \in \mathcal{F}} \left| \int_{\mathbb{R}^d} f(\boldsymbol{x}) p(\boldsymbol{x}) \, d\boldsymbol{x} - \int_{\mathbb{R}^d} f(\boldsymbol{x}) q(\boldsymbol{x}) \, d\boldsymbol{x} \right|, \tag{3}$$

  where $\mathcal{F}$ is a class of real-valued functions.

- For $\gamma_{\mathcal{F}}(p, q)$ to be a valid metric, the function class $\mathcal{F}$ must be sufficiently rich (e.g., the set of all bounded continuous functions).

Integral Probability Metric

- For any non-empty function class $\mathcal{F}$, $\gamma_{\mathcal{F}}(p, q)$ satisfies three properties of a distance metric:
  1. **Non-negativity**: $\gamma_{\mathcal{F}}(p, q) \geq 0$ and $\gamma_{\mathcal{F}}(p, p) = 0$.
  2. **Symmetry**: $\gamma_{\mathcal{F}}(p, q) = \gamma_{\mathcal{F}}(q, p)$.
  3. **Triangle Inequality**: $\gamma_{\mathcal{F}}(p, q) \leq \gamma_{\mathcal{F}}(p, r) + \gamma_{\mathcal{F}}(r, q)$ for any density r.
- Different choices of the function class $\mathcal{F}$ yield different statistical distances, including the total variation distance, the Earth mover's distance, and the maximum mean discrepancy.

Integral Probability Metric

Exercise 1

Let $p(\boldsymbol{x})$, $q(\boldsymbol{x})$, and $r(\boldsymbol{x})$ be probability density functions on $\mathcal{X}$, and let

$$\gamma_{\mathcal{F}}(p, q) := \sup_{f \in \mathcal{F}} \left| \int_{\mathcal{X}} f(\boldsymbol{x}) p(\boldsymbol{x}) \, d\boldsymbol{x} - \int_{\mathcal{X}} f(\boldsymbol{x}) q(\boldsymbol{x}) \, d\boldsymbol{x} \right|. \tag{4}$$

Show that $\gamma_{\mathcal{F}}$ satisfies the triangle inequality:

$$\gamma_{\mathcal{F}}(p, q) \leq \gamma_{\mathcal{F}}(p, r) + \gamma_{\mathcal{F}}(r, q). \tag{5}$$

Integral Probability Metric



Integral Probability Metric

- **Total Variation Distance**: The total variation distance, $\delta_{\mathrm{TV}}(p, q) := \frac{1}{2} \int_{\mathbb{R}^d} \left| p(\boldsymbol{x}) - q(\boldsymbol{x}) \right| d\boldsymbol{x}$, has an alternative expression as an IPM:

$$\delta_{\mathrm{TV}}(p, q) = \sup_{f \in \mathcal{F}_{\mathrm{ind}}} \left| \int_{\mathbb{R}^d} f(\boldsymbol{x}) p(\boldsymbol{x})\, d\boldsymbol{x} - \int_{\mathbb{R}^d} f(\boldsymbol{x}) q(\boldsymbol{x})\, d\boldsymbol{x} \right|, \tag{6}$$

  where $\mathcal{F}_{\mathrm{ind}} = \{f : f(\boldsymbol{x}) = I(\boldsymbol{x} \in A) \text{ for some } A \in \mathcal{A}\}$ is the class of indicator functions for all (measurable) events.
- **Earth Mover's Distance**: When $\mathcal{F}$ consists of all 1-Lipschitz continuous functions, the corresponding IPM becomes the Earth mover's distance (or 1-Wasserstein distance). Details will be discussed in Chapter 10.

Integral Probability Metric

- Why does $\delta_{\mathrm{TV}}(p, q) = \frac{1}{2} \int_{\mathbb{R}^d} |p(\boldsymbol{x}) - q(\boldsymbol{x})| \, d\boldsymbol{x}$ equal $\sup_{A \in \mathcal{A}} |P(A) - Q(A)|$?
- Define the set $A^* := \{\boldsymbol{x} \in \mathbb{R}^d : p(\boldsymbol{x}) \geq q(\boldsymbol{x})\}$. Since $\int_{\mathbb{R}^d} (p(\boldsymbol{x}) - q(\boldsymbol{x})) \, d\boldsymbol{x} = 1 - 1 = 0$, we have:

$$\int_{A^*} \big(p(\boldsymbol{x}) - q(\boldsymbol{x})\big) \, d\boldsymbol{x} = \int_{(A^*)^c} \big(q(\boldsymbol{x}) - p(\boldsymbol{x})\big) \, d\boldsymbol{x}. \tag{7}$$

- Consequently, the total variation distance can be written as:

$$\delta_{\mathrm{TV}}(p, q) = \frac{1}{2} \left(\int_{A^*} (p - q) \, d\boldsymbol{x} + \int_{(A^*)^c} (q - p) \, d\boldsymbol{x} \right) = \int_{A^*} \big(p(\boldsymbol{x}) - q(\boldsymbol{x})\big) \, d\boldsymbol{x}, \tag{8}$$

which attains the supremum $\sup_{A \in \mathcal{A}} \left| \int_A p(\boldsymbol{x}) \, d\boldsymbol{x} - \int_A q(\boldsymbol{x}) \, d\boldsymbol{x} \right|$ at $A = A^*$.

Integral Probability Metric

- We denote a kernel function by $k(\boldsymbol{x}, \boldsymbol{x}')$. For any symmetric positive definite k,[1] Mercer's theorem guarantees the existence of a kernel feature map φ such that[2]

$$k(\boldsymbol{x}, \boldsymbol{x}') = \langle \varphi(\boldsymbol{x}), \varphi(\boldsymbol{x}') \rangle. \tag{9}$$

- For example, when $d = 1$, the quadratic kernel $k_{\text{quad}}(x, x') = (xx' + c)^2$ $(c > 0)$ corresponds to the feature map:

$$\varphi(x) = \left(c, \sqrt{2c}\, x, x^2\right)^\top \in \mathbb{R}^3, \tag{10}$$

which captures the first- and second-order monomials of x.

[1] A symmetric kernel k is called *positive definite* if $\sum_{i=1}^n \sum_{j=1}^n a_i a_j k(\boldsymbol{x}_i, \boldsymbol{x}_j) \geq 0$ for any $n \in \mathbb{N}$, $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^d$, and $a_1, \ldots, a_n \in \mathbb{R}$.

[2] The symbol ϕ is more commonly used, but we use φ here to avoid confusion with the parameters of auxiliary networks.

Integral Probability Metric

Exercise 2

Let $c > 0$ be a constant, and define the univariate feature map $\varphi : \mathbb{R} \to \mathbb{R}^3$ and the quadratic kernel $k_{\text{quad}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ by

$$\varphi(x) = \left(c, \sqrt{2c}\, x, x^2\right)^\top \quad \text{and} \quad k_{\text{quad}}(x, x') = (xx' + c)^2. \tag{11}$$

1. Verify directly that the inner product $\varphi(x)^\top \varphi(x')$ equals $k_{\text{quad}}(x, x')$ for all $x, x' \in \mathbb{R}$.
2. Let X be a random variable with mean $\mu = E[X]$ and variance $\sigma^2 = \text{Var}[X]$. Express the expected feature vector $E[\varphi(X)]$ in terms of c, μ, and σ^2.

Integral Probability Metric



Integral Probability Metric

- As another example, for the radial basis function (RBF) kernel $k_{\text{RBF}}(x, x') = \exp\left(-\frac{(x-x')^2}{2\sigma^2}\right)$ when $d = 1$, expanding $\exp\left(\frac{xx'}{\sigma^2}\right) = \sum_{m=0}^{\infty} \frac{(xx')^m}{\sigma^{2m} m!}$ yields the feature map:

$$\varphi(x) = \exp\left(-\frac{x^2}{2\sigma^2}\right)\left(1, \frac{x}{\sigma\sqrt{1!}}, \frac{x^2}{\sigma^2\sqrt{2!}}, \dots\right)^\top. \tag{12}$$

- For the RBF kernel, two distributions p and q have the same expected feature map, $\int \varphi(\boldsymbol{x}) p(\boldsymbol{x})\, d\boldsymbol{x} = \int \varphi(\boldsymbol{x}) q(\boldsymbol{x})\, d\boldsymbol{x}$, if and only if $p = q$ (Gretton et al., 2012).

Integral Probability Metric

- **Maximum Mean Discrepancy (MMD)**: The MMD (Gretton et al., 2012) between $p(\boldsymbol{x})$ and $q(\boldsymbol{x})$ can be expressed as[3]

$$\mathrm{MMD}_k(p, q) = \left\| \int_{\mathbb{R}^d} \varphi(\boldsymbol{x}) p(\boldsymbol{x})\, d\boldsymbol{x} - \int_{\mathbb{R}^d} \varphi(\boldsymbol{x}) q(\boldsymbol{x})\, d\boldsymbol{x} \right\|_{\mathcal{H}}. \tag{13}$$

- Letting $\boldsymbol{\mu}_p := \int_{\mathbb{R}^d} \varphi(\boldsymbol{x}) p(\boldsymbol{x})\, d\boldsymbol{x}$ and $\boldsymbol{\mu}_q := \int_{\mathbb{R}^d} \varphi(\boldsymbol{x}) q(\boldsymbol{x})\, d\boldsymbol{x}$, it is known that the squared RHS reduces to the standard inner product form:

$$\mathrm{MMD}_k^2(p, q) = \langle \boldsymbol{\mu}_p - \boldsymbol{\mu}_q, \boldsymbol{\mu}_p - \boldsymbol{\mu}_q \rangle = (\boldsymbol{\mu}_p - \boldsymbol{\mu}_q)^\top (\boldsymbol{\mu}_p - \boldsymbol{\mu}_q). \tag{14}$$

- MMD builds a kernel-based statistic for a two-sample test:

$$H_0 : p = q \quad \text{vs.} \quad H_1 : p \neq q. \tag{15}$$

[3] The MMD is defined as the distance between kernel means in the feature space $\mathcal{H}$.

Integral Probability Metric

- Using $k(\boldsymbol{x}, \boldsymbol{x}') = \langle \varphi(\boldsymbol{x}), \varphi(\boldsymbol{x}') \rangle$, the squared MMD can be expanded in terms of the kernel function k as:

$$\begin{aligned} \mathrm{MMD}_k^2(p, q) = &\int_{\mathbb{R}^d \times \mathbb{R}^d} k(\boldsymbol{x}, \boldsymbol{x}') p(\boldsymbol{x}) p(\boldsymbol{x}') \, d\boldsymbol{x} \, d\boldsymbol{x}' - 2 \int_{\mathbb{R}^d \times \mathbb{R}^d} k(\boldsymbol{x}, \boldsymbol{y}) p(\boldsymbol{x}) q(\boldsymbol{y}) \, d\boldsymbol{x} \, d\boldsymbol{y} \\ &+ \int_{\mathbb{R}^d \times \mathbb{R}^d} k(\boldsymbol{y}, \boldsymbol{y}') q(\boldsymbol{y}) q(\boldsymbol{y}') \, d\boldsymbol{y} \, d\boldsymbol{y}'. \end{aligned} \tag{16}$$

- This kernel representation allows us to compute $\mathrm{MMD}_k^2(p, q)$ directly using $k(\cdot, \cdot)$, even when the feature map $\varphi(\boldsymbol{x})$ is infinite-dimensional (as in the RBF kernel).

Integral Probability Metric

Exercise 3

Let $p(\boldsymbol{x})$ and $q(\boldsymbol{y})$ be two probability density functions on $\mathbb{R}^d$, and let $k(\boldsymbol{x}, \boldsymbol{x}') = \varphi(\boldsymbol{x})^\top \varphi(\boldsymbol{x}')$ be a kernel function with feature map $\varphi : \mathbb{R}^d \to \mathbb{R}^m$. Define the kernel means $\boldsymbol{\mu}_p := \mathrm{E}_{\boldsymbol{X}\sim p}[\varphi(\boldsymbol{X})]$ and $\boldsymbol{\mu}_q := \mathrm{E}_{\boldsymbol{Y}\sim q}[\varphi(\boldsymbol{Y})]$, and define $\mathrm{MMD}_k^2(p, q) := (\boldsymbol{\mu}_p - \boldsymbol{\mu}_q)^\top (\boldsymbol{\mu}_p - \boldsymbol{\mu}_q)$. Show that for mutually independent random vectors $\boldsymbol{X}, \boldsymbol{X}' \overset{\text{i.i.d.}}{\sim} p$ and $\boldsymbol{Y}, \boldsymbol{Y}' \overset{\text{i.i.d.}}{\sim} q$:

$$\begin{aligned}\mathrm{MMD}_k^2(p, q) = \mathrm{E}_{(\boldsymbol{X},\boldsymbol{X}')\sim p(\boldsymbol{x})p(\boldsymbol{x}')}\left[k(\boldsymbol{X}, \boldsymbol{X}')\right] - 2\,\mathrm{E}_{(\boldsymbol{X},\boldsymbol{Y})\sim p(\boldsymbol{x})q(\boldsymbol{y})}\left[k(\boldsymbol{X}, \boldsymbol{Y})\right] \\ + \mathrm{E}_{(\boldsymbol{Y},\boldsymbol{Y}')\sim q(\boldsymbol{y})q(\boldsymbol{y}')}\left[k(\boldsymbol{Y}, \boldsymbol{Y}')\right].\end{aligned} \tag{17}$$

Integral Probability Metric



Generative Moment Matching Network

- Generative Moment Matching Networks (GMMNs) (Li et al., 2015) use an empirical MMD as a loss to train generative models, avoiding the need for an adversarial network as in GANs.
- Given minibatch samples $(\boldsymbol{x}_i)_{i=1}^B \overset{\text{i.i.d.}}{\sim} p(\boldsymbol{x})$ and $(G_\theta(\boldsymbol{z}_i))_{i=1}^B \overset{\text{i.i.d.}}{\sim} p_\theta(\boldsymbol{x})$, the minibatch-based empirical estimator for $\text{MMD}_k^2(p, p_\theta)$ can be expressed as:

$$\begin{aligned}\widehat{\text{MMD}}_k^2(p, p_\theta) = &\frac{1}{B(B-1)} \sum_{i=1}^{B} \sum_{j \neq i}^{B} k(\boldsymbol{x}_i, \boldsymbol{x}_j) - \frac{2}{B^2} \sum_{i=1}^{B} \sum_{j=1}^{B} k\big(\boldsymbol{x}_i, G_\theta(\boldsymbol{z}_j)\big) \\ &+ \frac{1}{B(B-1)} \sum_{i=1}^{B} \sum_{j \neq i}^{B} k\big(G_\theta(\boldsymbol{z}_i), G_\theta(\boldsymbol{z}_j)\big).\end{aligned} \tag{18}$$

Generative Moment Matching Network

- For any $i \neq j$, since $\boldsymbol{X}_i$ and $\boldsymbol{X}_j$ are independent draws from $p(\boldsymbol{x})$, $E[k(\boldsymbol{X}_i, \boldsymbol{X}_j)] = \int k(\boldsymbol{x}, \boldsymbol{x}')p(\boldsymbol{x})p(\boldsymbol{x}')\, d\boldsymbol{x}\, d\boldsymbol{x}'$.
- Consequently, by the linearity of expectation, $\widehat{\mathrm{MMD}}_k^2(p, p_\theta)$ is an unbiased estimator of $\mathrm{MMD}_k^2(p, p_\theta)$:

$$E\left[\widehat{\mathrm{MMD}}_k^2(p, p_\theta)\right] = \mathrm{MMD}_k^2(p, p_\theta). \tag{19}$$

- In practice, GMMNs use a mixture of multiple Gaussian RBF kernels with various bandwidth parameters σ_s to capture differences across multiple scales.

Generative Moment Matching Network

- Minimizing $\text{MMD}_k(p, p_\theta)$ can be interpreted as matching moments between p and p_θ.
- Let k be the kernel that defines the MMD, and let $\varphi(\boldsymbol{x})$ represent the corresponding kernel feature mapping:

$$k(\boldsymbol{x}, \boldsymbol{x}') = \varphi(\boldsymbol{x})^\top \varphi(\boldsymbol{x}'). \tag{20}$$

- Kernels with higher polynomial degrees (or the infinite-dimensional RBF kernel) allow for matching higher-order moments between the data and model distributions.

Generative Moment Matching Network

- When diagonal terms ($i = j$) are included in the minibatch sums, the minibatch-based empirical estimator for the squared MMD in GMMNs can be expressed as the squared distance between sample feature means:

$$\left\| \frac{1}{B} \sum_{i=1}^{B} \varphi(\boldsymbol{x}_i) - \frac{1}{B} \sum_{i=1}^{B} \varphi\big(G_\theta(\boldsymbol{z}_i)\big) \right\|^2 . \tag{21}$$

- Because $k(\boldsymbol{x}, G_\theta(\boldsymbol{z}))$ is a smooth function of $G_\theta(\boldsymbol{z})$, the generator parameters θ can be trained directly via standard backpropagation.

MMD GAN

- GMMNs face challenges in selecting effective kernels. MMD GANs (Li et al., 2017) overcome this limitation by introducing adversarial kernel learning.
- MMD GANs aim to target $\max_{k \in \mathcal{K}} \mathrm{MMD}_k(p, p_\theta)$, where $\mathcal{K}$ is a class of kernel functions.
- To model an expressive class $\mathcal{K}$, MMD GANs employ a neural network E_ϕ to define $(k \circ E_\phi)(\boldsymbol{x}, \boldsymbol{x}') := k\big(E_\phi(\boldsymbol{x}), E_\phi(\boldsymbol{x}')\big)$, targeting:

$$\max_{\phi} \mathrm{MMD}_{k \circ E_\phi}(p, p_\theta). \tag{22}$$

MMD GAN

- The injectivity (one-to-one property) of E_ϕ is crucial to retain the important properties of MMDs with usual kernels (i.e., $\text{MMD}_{k \circ E_\phi}(p, p_\theta) = 0 \iff p = p_\theta$).
- To enforce injectivity, MMD GANs incorporate an encoder architecture for E_ϕ, add a decoder network, and introduce a reconstruction error-based penalty term during training.

Other IPMs

- One of the main challenges in using general IPMs,

$$\gamma_{\mathcal{F}}(p, p_\theta) := \sup_{f \in \mathcal{F}} \left| \int_{\mathbb{R}^d} f(\boldsymbol{x}) p(\boldsymbol{x})\, d\boldsymbol{x} - \int_{\mathbb{R}^d} f(\boldsymbol{x}) p_\theta(\boldsymbol{x})\, d\boldsymbol{x} \right|, \tag{23}$$

  lies in approximating the supremum over the function class $\mathcal{F}$.
- While MMD has a tractable kernel representation that allows for the direct use of its empirical estimators, this is not the case for more general IPMs.
- Most methods targeting other IPMs employ neural networks to model elements within $\mathcal{F}$. Notably, Wasserstein GANs (Arjovsky et al., 2017) have become one of the most popular methods targeting the 1-Wasserstein distance.

Lecture Topics

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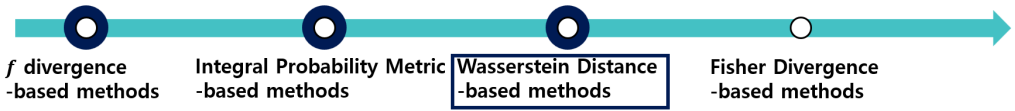
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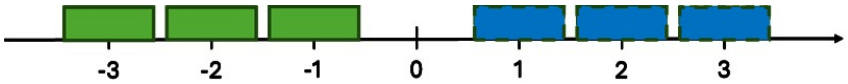
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Optimal Transport



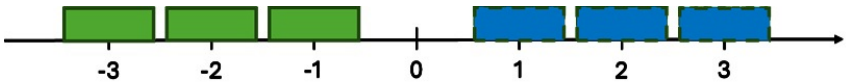
- Another line of work has targeted Wasserstein distances from an optimal transport perspective.
- Useful materials include:
  1. *Optimal Transport: Old and New* (Villani et al., 2009)
  2. *Optimal Transport for Applied Mathematicians* (Santambrogio, 2015)

Motivation: Monge Problem



- *Optimal Transport* is a problem focused on identifying maps that most efficiently move one distribution of mass to another and investigating their properties.
- Let's consider a toy example of moving green bold boxes on the left to blue dashed-box regions on the right.
- Which way is more efficient?
  1. T_1: $T_1(-3) = 1$, $T_1(-2) = 2$, and $T_1(-1) = 3$
  2. T_2: $T_2(-3) = 3$, $T_2(-2) = 2$, and $T_2(-1) = 1$

Motivation: Monge Problem



- The answer depends on criteria. When we consider the squared L_2-distance between the original green boxes and their transportation results:

$$\begin{aligned}&\left((-3-T_1(-3))^2+(-2-T_1(-2))^2+(-1-T_1(-1))^2\right)/3\\&=\left((-3-1)^2+(-2-2)^2+(-1-3)^2\right)/3=16\end{aligned} \tag{24}$$

$$\begin{aligned}&\left((-3-T_2(-3))^2+(-2-T_2(-2))^2+(-1-T_2(-1))^2\right)/3\\&=\left((-3-3)^2+(-2-2)^2+(-1-1)^2\right)/3=56/3\end{aligned} \tag{25}$$

Q: How many maps are there? What happens when we consider the L_1-distance?

Motivation: Monge Problem

Exercise 4

Let X be a discrete random variable uniformly distributed on $\mathcal{X}_S = \{-3, -2, -1\}$, so that $P(X = x) = 1/3$ for each $x \in \mathcal{X}_S$. Consider two transport maps $T_1, T_2 : \mathcal{X}_S \to \{1, 2, 3\}$ defined by:

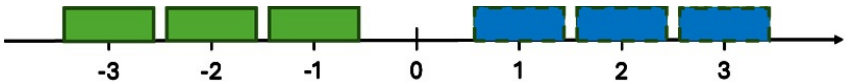
$$\begin{aligned} T_1(-3) &= 1, \quad T_1(-2) = 2, \quad T_1(-1) = 3, \\ T_2(-3) &= 3, \quad T_2(-2) = 2, \quad T_2(-1) = 1. \end{aligned} \tag{26}$$

Under the L_1 cost function $c(x, x') = |x - x'|$, compute and compare the expected transportation costs $E[c(X, T_1(X))] = \frac{1}{3}\sum_{x \in \mathcal{X}_S} |x - T_1(x)|$ and $E[c(X, T_2(X))] = \frac{1}{3}\sum_{x \in \mathcal{X}_S} |x - T_2(x)|$.

Motivation: Monge Problem



Motivation: Monge Problem



- **Monge Problem**: Given $p(\boldsymbol{x})$ and $q(\boldsymbol{x})$ and a cost function $c : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$, the optimal transportation cost is defined as:[4]

$$\inf_{T: T_{\#}p = q} \int_{\mathbb{R}^d} c(\boldsymbol{x}, T(\boldsymbol{x}))p(\boldsymbol{x})\, d\boldsymbol{x}, \tag{27}$$

where the constraint $T_{\#}p = q$ represents the push-forward operation, i.e., $p(\boldsymbol{x}) = q(T(\boldsymbol{x}))|\det J_T(\boldsymbol{x})|$. An optimal solution T^* to this problem, if it exists, is referred to as an *optimal transport map*.

[4] Optimal transport is more generally formulated with probability measures, but for simplicity, we state definitions using PDFs in this course.

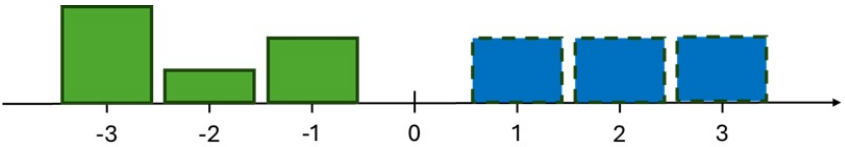
Motivation: Monge Problem

- When the cost function is the squared L_2-distance, $c(\boldsymbol{x}, \boldsymbol{x}') = \|\boldsymbol{x} - \boldsymbol{x}'\|^2$, T^* exists and is unique.
- In the univariate case ($d = 1$), the transport map T^* that moves percentiles of $p(x)$ to the corresponding percentiles of $q(x)$ is optimal:

$$T^*(x) = F_q^{-1}\big(F_p(x)\big), \tag{28}$$

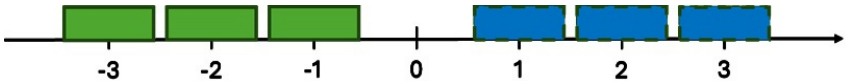
where F_p and F_q denote the CDFs of p and q, respectively.

Motivation: Kantorovich Problem



- The constraint on the optimal transport map, $T_{\#}p = q$, generally poses challenges in defining the optimal transportation cost.
- The *Kantorovich Problem* is a generalization of the Monge Problem to alleviate these challenges.

Motivation: Kantorovich Problem

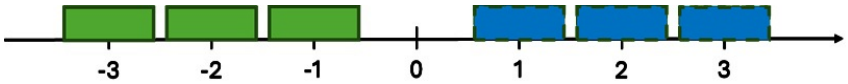


- In the first example, the joint distribution of $(x, T_1(x))$ can be expressed as:

(Source, Target)	1	2	3	Total
-3	1/3	0	0	1/3
-2	0	1/3	0	1/3
-1	0	0	1/3	1/3
Total	1/3	1/3	1/3	1

- That is, each (deterministic) transport map identifies the corresponding joint distribution of the source data and their transported results.

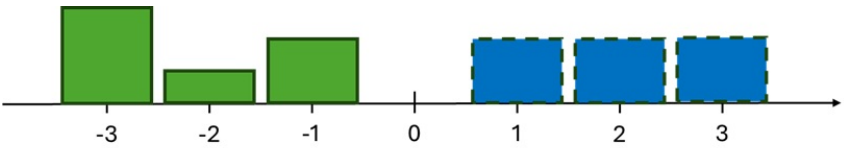
Motivation: Kantorovich Problem



- The Kantorovich Problem involves minimizing transportation costs among all joint distributions whose marginals are the source and target data distributions:

(Source, Target)	1	2	3	Total
-3	$p_{(-3,1)}$	$p_{(-3,2)}$	$p_{(-3,3)}$	1/3
-2	$p_{(-2,1)}$	$p_{(-2,2)}$	$p_{(-2,3)}$	1/3
-1	$p_{(-1,1)}$	$p_{(-1,2)}$	$p_{(-1,3)}$	1/3
Total	1/3	1/3	1/3	1

Motivation: Kantorovich Problem



(Source, Target)	1	2	3	Total
-3	$p_{(-3,1)}$	$p_{(-3,2)}$	$p_{(-3,3)}$	1/2
-2	$p_{(-2,1)}$	$p_{(-2,2)}$	$p_{(-2,3)}$	1/6
-1	$p_{(-1,1)}$	$p_{(-1,2)}$	$p_{(-1,3)}$	1/3
Total	1/3	1/3	1/3	1

Motivation: Kantorovich Problem

$$\min_{p_{(-3,1)},\cdots,p_{(-1,3)}} \Big(c(-3,1)p_{(-3,1)} + \cdots + c(-1,3)p_{(-1,3)} \Big)$$

subject to

$$\begin{aligned}
&p_{(-3,1)} + p_{(-3,2)} + p_{(-3,3)} = 1/2, \\
&p_{(-2,1)} + p_{(-2,2)} + p_{(-2,3)} = 1/6, \\
&p_{(-1,1)} + p_{(-1,2)} + p_{(-1,3)} = 1/3, \\
&p_{(-3,1)} + p_{(-2,1)} + p_{(-1,1)} = 1/3, \\
&p_{(-3,2)} + p_{(-2,2)} + p_{(-1,2)} = 1/3, \\
&p_{(-3,3)} + p_{(-2,3)} + p_{(-1,3)} = 1/3, \\
&p_{(-3,1)} \geq 0, \ldots, p_{(-1,3)} \geq 0.
\end{aligned} \tag{29}$$

Motivation: Kantorovich Problem

Exercise 5

Consider a discrete optimal transport problem from a source distribution on $\mathcal{X}_S = \{-3, -2, -1\}$ with marginal probabilities $(p_S(-3), p_S(-2), p_S(-1)) = (1/2, 1/6, 1/3)$ to a target distribution on $\mathcal{X}_T = \{1, 2, 3\}$ with marginal probabilities $(p_T(1), p_T(2), p_T(3)) = (1/3, 1/3, 1/3)$. Suppose a candidate joint PMF $p_{(i,j)}$ on $\mathcal{X}_S \times \mathcal{X}_T$ is given by

$$p_{(-3,1)} = \frac{1}{3}, \quad p_{(-3,2)} = \frac{1}{6}, \quad p_{(-2,2)} = \frac{1}{6}, \quad p_{(-1,3)} = \frac{1}{3}, \tag{30}$$

and $p_{(i,j)} = 0$ for all other pairs $(i, j) \in \mathcal{X}_S \times \mathcal{X}_T$.

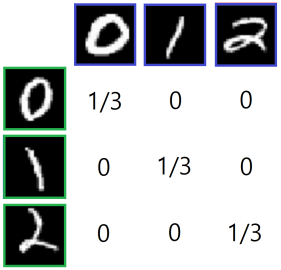
Motivation: Kantorovich Problem

1. Verify that $\sum_{j \in \mathcal{X}_T} p_{(i,j)} = p_S(i)$ for all $i \in \mathcal{X}_S$ and $\sum_{i \in \mathcal{X}_S} p_{(i,j)} = p_T(j)$ for all $j \in \mathcal{X}_T$.

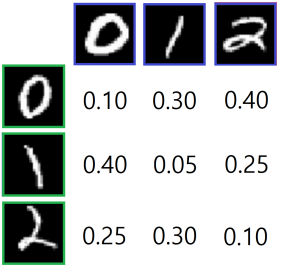
Motivation: Kantorovich Problem

2. Compute the expected squared L_2 transportation cost $\sum_{i \in \mathcal{X}_S} \sum_{j \in \mathcal{X}_T} (i-j)^2 p_{(i,j)}$.

Application to High-dimensional Data



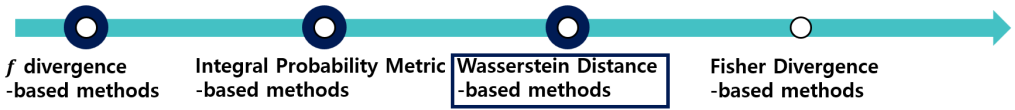
Joint Distribution



Transportation Cost

$$(0.10 \times (1/3) + 0.30 \times 0 + 0.40 \times 0)$$
$$+ (0.40 \times 0 + 0.05 \times (1/3) + 0.25 \times 0)$$
$$+ (0.25 \times 0 + 0.30 \times 0 + 0.10 \times (1/3)) = 0.08$$

Wasserstein Distance



- The p-**Wasserstein distance** is defined as:

$$W_p(p, q) := \left(\inf_{\pi \in \Pi(p,q)} \int_{\mathbb{R}^d \times \mathbb{R}^d} d^p(\boldsymbol{x}, \boldsymbol{x}')\, \pi(\boldsymbol{x}, \boldsymbol{x}')\, d\boldsymbol{x}\, d\boldsymbol{x}' \right)^{1/p}, \tag{31}$$

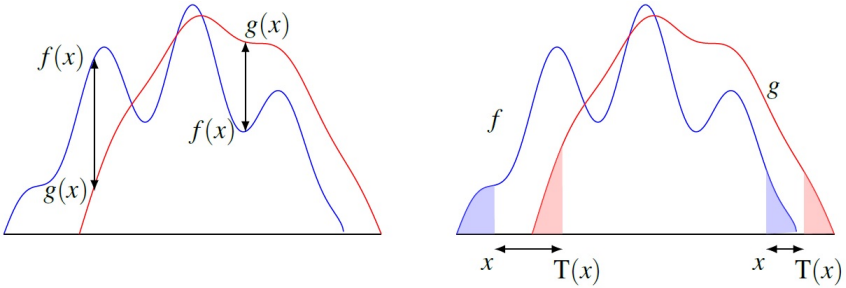
where $p \in [1, \infty)$, d is a metric defined on $\mathbb{R}^d$,[5] and $\Pi(p, q)$ denotes the set of all joint distributions (couplings) whose marginal densities are p and q.

[5] The Kantorovich problem is generally formulated using a general cost function $c(\boldsymbol{x}, \boldsymbol{x}')$.

Wasserstein Distance

- Under some regularity conditions, there exists an optimal transport map T^* that satisfies:
  1. $W_p(p, q) = \left(\int_{\mathbb{R}^d} d^p(\boldsymbol{x}, T^*(\boldsymbol{x})) p(\boldsymbol{x})\, d\boldsymbol{x} \right)^{1/p}$
  2. $T^*_{\#} p = q$
- That is, the Monge Problem can be equivalent to the Kantorovich Problem, and together they are simply referred to as the *Monge-Kantorovich Problem*.

Wasserstein Distance



- The f-divergence utilizes density-ratios to quantify discrepancies between distributions.
- In contrast, Wasserstein distances utilize transportation costs.

The figure is from Santambrogio (2015).

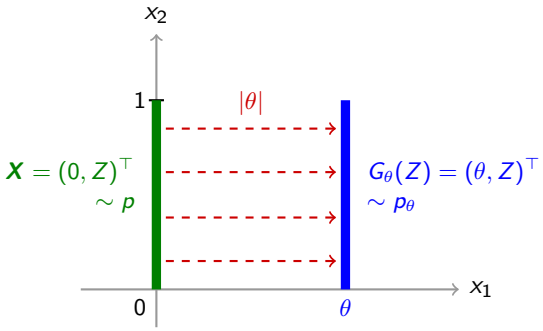
Wasserstein Distance

- For example, when d is the Euclidean norm, W_p becomes the Mallows metric (Mallows, 1972), and has played an important role in deriving asymptotic properties of bootstrap estimators (Bickel and Freedman, 1981; Freedman, 1981).
- Wasserstein distances effectively quantify differences between high-dimensional distributions when their supports are in low-dimensional manifolds:

Example (Example 1 in Arjovsky et al., 2017)

Let $Z \sim \text{Unif}(0,1)$, $\boldsymbol{X} = (0, Z)^\top \sim p$, and $G_\theta(Z) = (\theta, Z)^\top \sim p_\theta$ in $\mathbb{R}^2$.

Wasserstein Distance



- The real distribution p is supported on the vertical line segment $\{0\} \times [0, 1]$, while p_θ is supported on $\{\theta\} \times [0, 1]$ in $\mathbb{R}^2$.
- Whenever $\theta \neq 0$, the two supports are completely disjoint, so the density ratio $p(\boldsymbol{x})/p_\theta(\boldsymbol{x})$ is degenerate everywhere.
- In contrast, optimal transport moves each point $(0, z)$ horizontally to (θ, z) over a Euclidean distance of $|\theta|$.

Wasserstein Distance

Exercise 6

Let $Z \sim \text{Unif}(0,1)$, and define two random vectors $\boldsymbol{X} = (0, Z)^\top \sim p$ and $\boldsymbol{X}_\theta = (\theta, Z)^\top \sim p_\theta$ in $\mathbb{R}^2$ for a parameter $\theta \in \mathbb{R}$. Consider the transport map $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $T(\boldsymbol{x}) = (x_1 + \theta,\, x_2)^\top$ for $\boldsymbol{x} = (x_1, x_2)^\top$.

1. Verify that $T(\boldsymbol{X}) = \boldsymbol{X}_\theta$.
2. Let $d(\boldsymbol{x}, \boldsymbol{x}') = \|\boldsymbol{x} - \boldsymbol{x}'\|_2 = \sqrt{(x_1 - x_1')^2 + (x_2 - x_2')^2}$ be the Euclidean distance on $\mathbb{R}^2$. For any $p \geq 1$, compute the transportation cost $\left(\mathrm{E}\left[d^p(\boldsymbol{X}, T(\boldsymbol{X}))\right]\right)^{1/p}$.

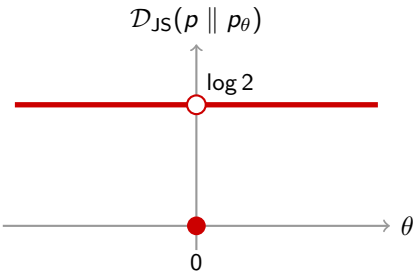
Wasserstein Distance

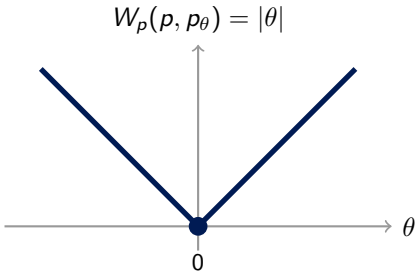


Wasserstein Distance

- In Example 1 of Arjovsky et al. (2017) ($\boldsymbol{X} = (0, Z)^\top \sim p$ and $G_\theta(Z) = (\theta, Z)^\top \sim p_\theta$ with $Z \sim \text{Unif}(0, 1)$), intuitively, the statistical distance between p and p_θ should decrease continuously as θ vanishes:
  - $W_p(p, p_\theta) = |\theta|$
  - $\mathcal{D}_{\text{JS}}(p \parallel p_\theta) = \log 2$ if $\theta \neq 0$, and 0 if $\theta = 0$
  - $\mathcal{D}_{\text{KL}}(p \parallel p_\theta) = \infty$ if $\theta \neq 0$, and 0 if $\theta = 0$
  - $\delta_{\text{TV}}(p, p_\theta) = 1$ if $\theta \neq 0$, and 0 if $\theta = 0$

Wasserstein Distance





- Because $\mathcal{D}_{\mathrm{JS}}(p \parallel p_\theta)$ and $\delta_{\mathrm{TV}}(p, p_\theta)$ are constant for all $\theta \neq 0$, their gradients with respect to θ vanish, providing no learning signal for gradient descent.
- In contrast, $W_p(p, p_\theta) = |\theta|$ is continuous and provides an informative gradient everywhere to push $\theta \to 0$.

1-Wasserstein Distance: Duality

- When $p = 1$ (often called the *Earth Mover's Distance*), the Kantorovich-Rubinstein duality holds (Villani et al., 2009; Villani, 2021):

$$W_1(p, q) = \sup_{\substack{f,g: \\ f(\boldsymbol{x})+g(\boldsymbol{x}')\leq d(\boldsymbol{x},\boldsymbol{x}')}} \left(\int_{\mathbb{R}^d} f(\boldsymbol{x}) p(\boldsymbol{x})\, d\boldsymbol{x} + \int_{\mathbb{R}^d} g(\boldsymbol{x}') q(\boldsymbol{x}')\, d\boldsymbol{x}' \right). \quad (32)$$

- When $d(\boldsymbol{x}, \boldsymbol{x}') = \|\boldsymbol{x} - \boldsymbol{x}'\|$ for some norm $\|\cdot\|$, the dual form simplifies to:

$$W_1(p, q) = \sup_{\mathrm{Lip}(f)\leq 1} \left(\int_{\mathbb{R}^d} f(\boldsymbol{x}) p(\boldsymbol{x})\, d\boldsymbol{x} - \int_{\mathbb{R}^d} f(\boldsymbol{x}) q(\boldsymbol{x})\, d\boldsymbol{x} \right), \quad (33)$$

where $\mathrm{Lip}(f) := \sup_{\boldsymbol{x}\neq\boldsymbol{x}'} \frac{|f(\boldsymbol{x})-f(\boldsymbol{x}')|}{\|\boldsymbol{x}-\boldsymbol{x}'\|}$ denotes the Lipschitz constant of f. This formulation confirms that W_1 is an Integral Probability Metric (IPM).

Duality: Sketch of Proof

- For any $\pi(\boldsymbol{x}, \boldsymbol{x}') \geq 0$, note that

$$\sup_{f,g} \left(\int_{\mathbb{R}^d} f(\boldsymbol{x}) p(\boldsymbol{x})\, d\boldsymbol{x} + \int_{\mathbb{R}^d} g(\boldsymbol{x}') q(\boldsymbol{x}')\, d\boldsymbol{x}' - \int_{\mathbb{R}^d \times \mathbb{R}^d} \big(f(\boldsymbol{x}) + g(\boldsymbol{x}')\big) \pi(\boldsymbol{x}, \boldsymbol{x}')\, d\boldsymbol{x}\, d\boldsymbol{x}' \right)$$

equals 0 when $\pi \in \Pi(p, q)$ and $+\infty$ otherwise. Thus:

$$W_1(p, q) = \inf_{\pi \geq 0} \sup_{f,g} \Big(\int_{\mathbb{R}^d \times \mathbb{R}^d} d(\boldsymbol{x}, \boldsymbol{x}') \pi(\boldsymbol{x}, \boldsymbol{x}')\, d\boldsymbol{x}\, d\boldsymbol{x}' + \int_{\mathbb{R}^d} f(\boldsymbol{x}) p(\boldsymbol{x})\, d\boldsymbol{x} + \int_{\mathbb{R}^d} g(\boldsymbol{x}') q(\boldsymbol{x}')\, d\boldsymbol{x}' - \int_{\mathbb{R}^d \times \mathbb{R}^d} \big(f(\boldsymbol{x}) + g(\boldsymbol{x}')\big) \pi(\boldsymbol{x}, \boldsymbol{x}')\, d\boldsymbol{x}\, d\boldsymbol{x}' \Big).$$

Duality: Sketch of Proof

- Now, by exchanging the infimum and supremum:[6]

$$
\begin{aligned}
W_1(p,q) &= \sup_{f,g} \inf_{\pi \geq 0} \Big(\int_{\mathbb{R}^d \times \mathbb{R}^d} d(\boldsymbol{x}, \boldsymbol{x}') \pi(\boldsymbol{x}, \boldsymbol{x}') \, d\boldsymbol{x} \, d\boldsymbol{x}' + \int_{\mathbb{R}^d} f(\boldsymbol{x}) p(\boldsymbol{x}) \, d\boldsymbol{x} \\
&\qquad + \int_{\mathbb{R}^d} g(\boldsymbol{x}') q(\boldsymbol{x}') \, d\boldsymbol{x}' - \int_{\mathbb{R}^d \times \mathbb{R}^d} \big(f(\boldsymbol{x}) + g(\boldsymbol{x}') \big) \pi(\boldsymbol{x}, \boldsymbol{x}') \, d\boldsymbol{x} \, d\boldsymbol{x}' \Big) \\
&= \sup_{f,g} \Big(\int_{\mathbb{R}^d} f(\boldsymbol{x}) p(\boldsymbol{x}) \, d\boldsymbol{x} + \int_{\mathbb{R}^d} g(\boldsymbol{x}') q(\boldsymbol{x}') \, d\boldsymbol{x}' \\
&\qquad + \inf_{\pi \geq 0} \int_{\mathbb{R}^d \times \mathbb{R}^d} \big(d(\boldsymbol{x}, \boldsymbol{x}') - f(\boldsymbol{x}) - g(\boldsymbol{x}') \big) \pi(\boldsymbol{x}, \boldsymbol{x}') \, d\boldsymbol{x} \, d\boldsymbol{x}' \Big).
\end{aligned}
$$

[6] See Section 1.2 in Santambrogio (2015) for details on conditions to exchange them.

Duality: Sketch of Proof

- The inner infimum over $\pi \geq 0$ is 0 if $f(\boldsymbol{x}) + g(\boldsymbol{x}') \leq d(\boldsymbol{x}, \boldsymbol{x}')$ everywhere and $-\infty$ otherwise, yielding:

$$W_1(p, q) = \sup_{\substack{f,g: \\ f(\boldsymbol{x})+g(\boldsymbol{x}') \leq d(\boldsymbol{x},\boldsymbol{x}')}} \left(\int_{\mathbb{R}^d} f(\boldsymbol{x}) p(\boldsymbol{x}) \, d\boldsymbol{x} + \int_{\mathbb{R}^d} g(\boldsymbol{x}') q(\boldsymbol{x}') \, d\boldsymbol{x}' \right). \quad (34)$$

- Next, assume $d(\boldsymbol{x}, \boldsymbol{x}') = \|\boldsymbol{x} - \boldsymbol{x}'\|$. For any feasible (f, g), define:

$$f^d(\boldsymbol{x}) := \inf_{\boldsymbol{x}'} \big(d(\boldsymbol{x}, \boldsymbol{x}') - g(\boldsymbol{x}') \big). \quad (35)$$

Then $f(\boldsymbol{x}) \leq f^d(\boldsymbol{x}) \leq d(\boldsymbol{x}, \boldsymbol{x}) - g(\boldsymbol{x}) = -g(\boldsymbol{x})$, which implies $\int fp + \int gq \leq \int f^d p - \int f^d q$.

Duality: Sketch of Proof

- Here, f^d is 1-Lipschitz continuous because by the triangle inequality:

$$f^d(\boldsymbol{x}_1) \leq \inf_{\boldsymbol{x}'} \left(d(\boldsymbol{x}_1, \boldsymbol{x}_2) + d(\boldsymbol{x}_2, \boldsymbol{x}') - g(\boldsymbol{x}') \right) = d(\boldsymbol{x}_1, \boldsymbol{x}_2) + f^d(\boldsymbol{x}_2), \tag{36}$$

  and $d(\boldsymbol{x}_1, \boldsymbol{x}_2) = d(\boldsymbol{x}_2, \boldsymbol{x}_1)$. This concludes the proof for LHS $\leq$ RHS.

- Conversely, for any 1-Lipschitz function f and any $\pi \in \Pi(p, q)$:

$$\begin{aligned} \int_{\mathbb{R}^d \times \mathbb{R}^d} d(\boldsymbol{x}, \boldsymbol{x}') \pi(\boldsymbol{x}, \boldsymbol{x}') \, d\boldsymbol{x} \, d\boldsymbol{x}' &\geq \int_{\mathbb{R}^d \times \mathbb{R}^d} \left(f(\boldsymbol{x}) - f(\boldsymbol{x}') \right) \pi(\boldsymbol{x}, \boldsymbol{x}') \, d\boldsymbol{x} \, d\boldsymbol{x}' \\ &= \int_{\mathbb{R}^d} f(\boldsymbol{x}) p(\boldsymbol{x}) \, d\boldsymbol{x} - \int_{\mathbb{R}^d} f(\boldsymbol{x}) q(\boldsymbol{x}) \, d\boldsymbol{x}. \end{aligned} \tag{37}$$

  Taking the infimum over $\pi \in \Pi(p, q)$ and supremum over $\mathrm{Lip}(f) \leq 1$ yields LHS $\geq$ RHS.

Primal vs. Dual Problems

- For general cases with cost function $c(\boldsymbol{x}, \boldsymbol{x}')$:
  1. **(Primal)**: $\inf_{\pi \in \Pi(p,q)} \int_{\mathbb{R}^d \times \mathbb{R}^d} c(\boldsymbol{x}, \boldsymbol{x}') \pi(\boldsymbol{x}, \boldsymbol{x}')\, d\boldsymbol{x}\, d\boldsymbol{x}'$
  2. **(Dual)**: $\sup_{f,g: f(\boldsymbol{x}) + g(\boldsymbol{x}') \le c(\boldsymbol{x}, \boldsymbol{x}')} \left(\int_{\mathbb{R}^d} f(\boldsymbol{x}) p(\boldsymbol{x})\, d\boldsymbol{x} + \int_{\mathbb{R}^d} g(\boldsymbol{x}') q(\boldsymbol{x}')\, d\boldsymbol{x}' \right)$
- When $d(\boldsymbol{x}, \boldsymbol{x}') = \|\boldsymbol{x} - \boldsymbol{x}'\|$:
  1. **(Primal)**: $W_1(p, q) = \inf_{\pi \in \Pi(p,q)} \int_{\mathbb{R}^d \times \mathbb{R}^d} d(\boldsymbol{x}, \boldsymbol{x}') \pi(\boldsymbol{x}, \boldsymbol{x}')\, d\boldsymbol{x}\, d\boldsymbol{x}'$
  2. **(Dual)**: $W_1(p, q) = \sup_{\mathrm{Lip}(f) \le 1} \left(\int_{\mathbb{R}^d} f(\boldsymbol{x}) p(\boldsymbol{x})\, d\boldsymbol{x} - \int_{\mathbb{R}^d} f(\boldsymbol{x}) q(\boldsymbol{x})\, d\boldsymbol{x} \right)$

Multi-marginal Transport Problem

- The optimal transportation cost is generally defined for multiple distributions $p^{(1)}, \ldots, p^{(M)}$ as follows:
  1. **(Primal)**:

$$\inf_{\pi \in \Pi(p^{(1)}, \ldots, p^{(M)})} \int c\left(\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(M)}\right) \pi\left(\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(M)}\right) d\boldsymbol{x}^{(1)} \cdots d\boldsymbol{x}^{(M)}, \tag{38}$$

  where $\Pi(p^{(1)}, \ldots, p^{(M)})$ represents the set of all joint distributions whose marginals are $p^{(1)}, \ldots, p^{(M)}$.
  2. **(Dual)**:

$$\sup_{\substack{f_1, \ldots, f_M: \\ \sum_{m=1}^{M} f_m(\boldsymbol{x}^{(m)}) \leq c(\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(M)})}} \sum_{m=1}^{M} \int_{\mathbb{R}^d} f_m\left(\boldsymbol{x}^{(m)}\right) p^{(m)}\left(\boldsymbol{x}^{(m)}\right) d\boldsymbol{x}^{(m)}. \tag{39}$$

Multi-marginal Transport Problem

Exercise 7

Consider $M = 3$ electrons orbiting a nucleus at the origin along the unit circle in $\mathbb{R}^2$. Each electron $m \in \{1, 2, 3\}$ has position $\boldsymbol{X}^{(m)} = (\cos\Theta_m, \sin\Theta_m)^\top$ with marginal $\Theta_m \sim \text{Unif}(0, 2\pi)$. Under the repulsive cost $c(\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \boldsymbol{x}^{(3)}) := -\sum_{1 \le i < j \le 3} \|\boldsymbol{x}^{(i)} - \boldsymbol{x}^{(j)}\|_2^2$:

1. Show that for any $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \boldsymbol{x}^{(3)}$ on the unit circle, $c\big(\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \boldsymbol{x}^{(3)}\big) \ge -9$.

Multi-marginal Transport Problem

2. Show that the equiangular rotation $\Theta_2 = \Theta_1 + \frac{2\pi}{3}$ and $\Theta_3 = \Theta_1 + \frac{4\pi}{3}$ achieves the minimum primal cost of -9.

Wasserstein Generative Adversarial Networks

- Wasserstein Generative Adversarial Networks (WGANs) (Arjovsky et al., 2017) parameterize 1-Lipschitz continuous functions using a neural network f_ϕ, optimizing:

$$\min_\theta \max_\phi \left(\int_{\mathbb{R}^d} f_\phi(\boldsymbol{x}) p_{\text{data}}(\boldsymbol{x})\, d\boldsymbol{x} - \int_{\mathbb{R}^d} f_\phi(\boldsymbol{x}) p_\theta(\boldsymbol{x})\, d\boldsymbol{x} \right). \tag{40}$$

- If the set $\{f_\phi \mid \phi \in \Phi\}$ perfectly captures all 1-Lipschitz functions $\{f \mid \text{Lip}(f) \leq 1\}$, Equation (40) equals $\min_\theta W_1(p_{\text{data}}, p_\theta)$.
- If the condition is relaxed to C-Lipschitz continuity, Equation (40) equals $C \cdot \min_\theta W_1(p_{\text{data}}, p_\theta)$.
- To enforce the Lipschitz constraint during training, WGANs clip the network weights to a compact interval $[-c, c]$.

Wasserstein Generative Adversarial Networks

Exercise 8

Let $p_{\text{data}}(\boldsymbol{x})$ and $p_\theta(\boldsymbol{x})$ be two probability density functions on $\mathbb{R}^d$. Recall that by the Kantorovich-Rubinstein duality, the 1-Wasserstein distance is given by

$$W_1(p_{\text{data}}, p_\theta) = \sup_{\text{Lip}(\tilde{f}) \leq 1} \left(\int_{\mathbb{R}^d} \tilde{f}(\boldsymbol{x}) p_{\text{data}}(\boldsymbol{x}) \, d\boldsymbol{x} - \int_{\mathbb{R}^d} \tilde{f}(\boldsymbol{x}) p_\theta(\boldsymbol{x}) \, d\boldsymbol{x} \right), \quad (41)$$

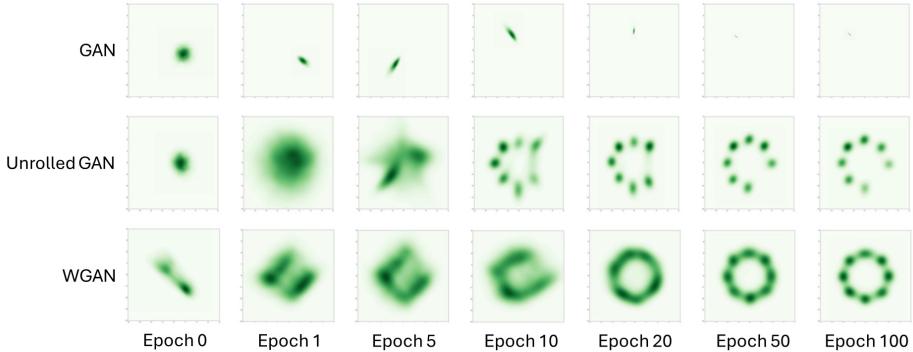
where $\text{Lip}(g) := \sup_{\boldsymbol{x} \neq \boldsymbol{x}'} \frac{|g(\boldsymbol{x}) - g(\boldsymbol{x}')|}{\|\boldsymbol{x} - \boldsymbol{x}'\|}$ denotes the Lipschitz constant of g. For any constant $C > 0$, show that:

$$\sup_{\text{Lip}(f) \leq C} \left(\int_{\mathbb{R}^d} f(\boldsymbol{x}) p_{\text{data}}(\boldsymbol{x}) \, d\boldsymbol{x} - \int_{\mathbb{R}^d} f(\boldsymbol{x}) p_\theta(\boldsymbol{x}) \, d\boldsymbol{x} \right) = C \cdot W_1(p_{\text{data}}, p_\theta). \quad (42)$$

Wasserstein Generative Adversarial Networks



Wasserstein Generative Adversarial Networks



- WGANs performed better than GAN baselines in learning the distribution of Gaussian mixtures with 8 modes.

Images are edited from Arjovsky et al. (2017).

WGAN with Gradient Penalty

- **WGAN with Gradient Penalty (WGAN-GP)**: Gulrajani et al. (2017) proposed a gradient-based penalty term to enforce the Lipschitz constraint in the dual form.
- For any $\pi \in \Pi(p_{\text{data}}, p_\theta)$ and $0 \le t \le 1$, let $\boldsymbol{X}^{(t)} := t\boldsymbol{X} + (1-t)\boldsymbol{X}'$ where $(\boldsymbol{X}, \boldsymbol{X}') \sim \pi$. The authors showed that the optimal 1-Lipschitz continuous function f^* satisfies:

$$\nabla_{\boldsymbol{X}^{(t)}} f^*(\boldsymbol{X}^{(t)}) = \frac{\boldsymbol{X}' - \boldsymbol{X}^{(t)}}{\|\boldsymbol{X}' - \boldsymbol{X}^{(t)}\|} \tag{43}$$

  for any norm $\|\cdot\|$.
- Motivated by this, the authors proposed to target the following objective:

$$\begin{aligned} &\int_{\mathbb{R}^d} f_\phi(\boldsymbol{x}) p_{\text{data}}(\boldsymbol{x})\, d\boldsymbol{x} - \int_{\mathbb{R}^d} f_\phi(\boldsymbol{x}) p_\theta(\boldsymbol{x})\, d\boldsymbol{x} \\ &\quad - \lambda\, \mathrm{E}_{\tilde{\boldsymbol{x}}}\left[\left(\|\nabla_{\tilde{\boldsymbol{x}}} f_\phi(\tilde{\boldsymbol{x}})\|_2 - 1 \right)^2 \right]. \end{aligned} \tag{44}$$

WGAN with Gradient Penalty

Exercise 9

Let $f^* : \mathbb{R}^d \to \mathbb{R}$ be a differentiable 1-Lipschitz function, i.e., $\|\nabla f^*(\boldsymbol{x})\|_2 \leq 1$ for all $\boldsymbol{x} \in \mathbb{R}^d$. For two distinct points $\boldsymbol{x}, \boldsymbol{x}' \in \mathbb{R}^d$ and $t \in (0,1)$, define $\boldsymbol{x}^{(t)} := t\boldsymbol{x} + (1-t)\boldsymbol{x}'$ and the unit vector $\boldsymbol{v} := \frac{\boldsymbol{x}' - \boldsymbol{x}^{(t)}}{\|\boldsymbol{x}' - \boldsymbol{x}^{(t)}\|_2}$. Suppose f^* attains the 1-Lipschitz bound along the line segment between $\boldsymbol{x}$ and $\boldsymbol{x}'$, satisfying $f^*(\boldsymbol{x}^{(t)} + h\boldsymbol{v}) = f^*(\boldsymbol{x}^{(t)}) + h$ for small $h \in \mathbb{R}$.

1. Show that $\boldsymbol{v}^\top \nabla f^*(\boldsymbol{x}^{(t)}) = 1$.

WGAN with Gradient Penalty

2. Show that

$$\nabla f^*(\boldsymbol{x}^{(t)}) = \frac{\boldsymbol{x}' - \boldsymbol{x}^{(t)}}{\|\boldsymbol{x}' - \boldsymbol{x}^{(t)}\|_2}. \tag{45}$$

Wasserstein Autoencoder

- **Wasserstein Autoencoder (WAE)**: Tolstikhin et al. (2018) derived an alternative representation of the p-Wasserstein distance, and proposed WAEs based on the result.
- Let $\Pi^\dagger(p_{\text{data}}, p_\theta; p_Z)$ be the set of $\pi^\dagger(\boldsymbol{x}, \boldsymbol{x}', \boldsymbol{z})$ such that $\pi^\dagger(\boldsymbol{x}) = p_{\text{data}}(\boldsymbol{x})$ and $\pi^\dagger(\boldsymbol{x}', \boldsymbol{z}) = p_\theta(\boldsymbol{x}' \mid \boldsymbol{z})p(\boldsymbol{z})$. Then:

$$\begin{aligned} &\inf_{\pi \in \Pi(p_{\text{data}}, p_\theta)} \int d^p(\boldsymbol{x}, \boldsymbol{x}')\pi(\boldsymbol{x}, \boldsymbol{x}')\, d\boldsymbol{x}\, d\boldsymbol{x}' \\ &\quad \leq \inf_{\pi^\dagger \in \Pi^\dagger(p_{\text{data}}, p_\theta; p_Z)} \int d^p(\boldsymbol{x}, \boldsymbol{x}')\pi^\dagger(\boldsymbol{x}, \boldsymbol{x}', \boldsymbol{z})\, d\boldsymbol{x}\, d\boldsymbol{x}'\, d\boldsymbol{z}. \end{aligned} \tag{46}$$

- When the generator (or decoder) is deterministic, i.e., $p_\theta(\boldsymbol{x}' \mid \boldsymbol{z})$ is a point mass at $G_\theta(\boldsymbol{z})$, equality holds and the RHS can be re-expressed as:

$$W_p^p(p_{\text{data}}, p_\theta) = \inf_{\substack{q(\boldsymbol{z}|\boldsymbol{x}): \\ \int q(\boldsymbol{z}|\boldsymbol{x})p_{\text{data}}(\boldsymbol{x})d\boldsymbol{x} = p(\boldsymbol{z})}} \int_{\mathbb{R}^d} \int_{\mathcal{Z}} d^p\big(\boldsymbol{x}, G_\theta(\boldsymbol{z})\big) q(\boldsymbol{z} \mid \boldsymbol{x}) p_{\text{data}}(\boldsymbol{x})\, d\boldsymbol{z}\, d\boldsymbol{x}.$$

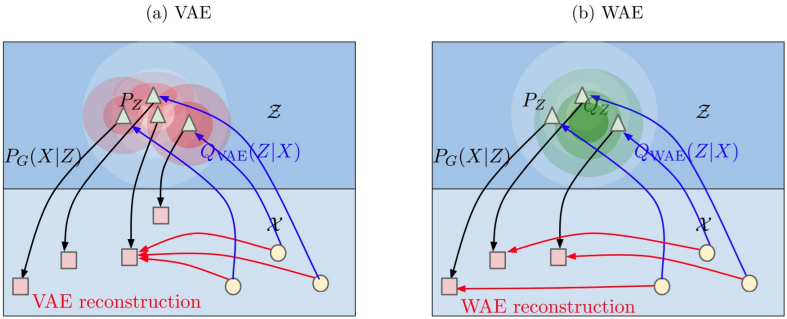
Wasserstein Autoencoder

- Based on this relation, WAEs introduce encoders $q_\phi(\boldsymbol{z} \mid \boldsymbol{x})$ and a penalty term to enforce the marginal constraint $\int_{\mathbb{R}^d} q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) p_{\text{data}}(\boldsymbol{x})\, d\boldsymbol{x} = p(\boldsymbol{z})$.
- The WAE loss can be expressed as:

$$\begin{aligned} \mathcal{L}_{\text{WAE}}(\theta, \phi) = &\int_{\mathbb{R}^d} \int_{\mathcal{Z}} d^p\big(\boldsymbol{x}, G_\theta(\boldsymbol{z})\big)\, q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) p_{\text{data}}(\boldsymbol{x})\, d\boldsymbol{z}\, d\boldsymbol{x} \\ &+ \lambda\, \mathcal{D}_{\boldsymbol{Z}} \left(\int_{\mathbb{R}^d} q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) p_{\text{data}}(\boldsymbol{x})\, d\boldsymbol{x},\, p(\boldsymbol{z}) \right), \end{aligned} \tag{47}$$

  where $\mathcal{D}_{\boldsymbol{Z}}$ indicates the statistical distance applied to the distributions of $\boldsymbol{Z}$.
- WAEs typically use JS divergence and MMD (Maximum Mean Discrepancy) as measures for $\mathcal{D}_{\boldsymbol{Z}}$.

Wasserstein Autoencoder



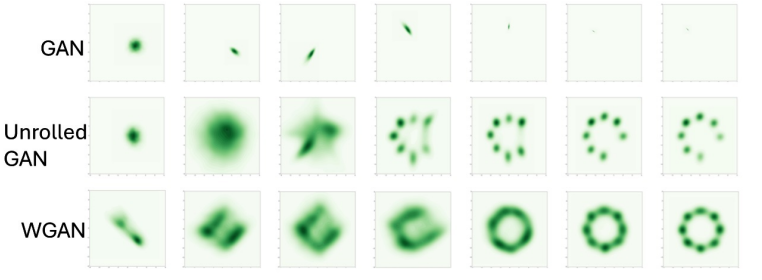
- Compared with the loss of VAEs (the negative ELBO), the penalty term changes from matching $q_\phi(\boldsymbol{z} \mid \boldsymbol{x})$ directly with $p(\boldsymbol{z})$ to matching $\int q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) p_{\text{data}}(\boldsymbol{x})\, d\boldsymbol{x}$ with $p(\boldsymbol{z})$.

The figure is from Tolstikhin et al. (2018).

Generation Results: WAEs and WGANs



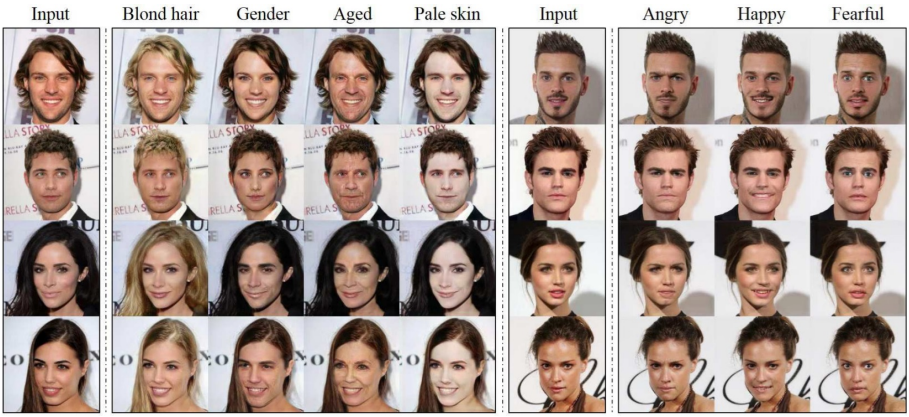
(a) Sharp generation results from WAEs



(b) Preventing mode collapse with WGANs

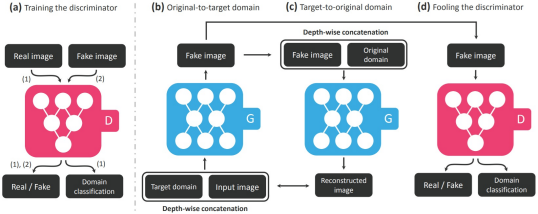
Images are edited from Arjovsky et al. (2017) and Tolstikhin et al. (2018).

Application: StarGAN



The figure is from Choi et al. (2018).

Application: StarGAN

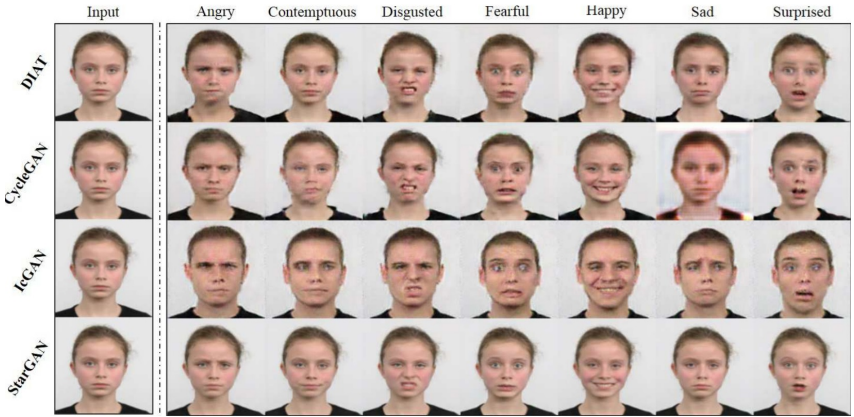


- Let $G_\theta(\cdot, \boldsymbol{c})$ be a translator that moves data from an arbitrary source domain to the target domain described by $\boldsymbol{c}$. For the adversarial loss component, StarGAN (Choi et al., 2018) targets:

$$\int f_\phi(\boldsymbol{x}) p_{\text{data}}(\boldsymbol{x})\, d\boldsymbol{x} - \int f_\phi\big(G_\theta(\boldsymbol{x}, \boldsymbol{c})\big) p_{\text{data}}(\boldsymbol{x}, \boldsymbol{c})\, d\boldsymbol{x}\, d\boldsymbol{c} \tag{48}$$

with an added gradient-penalty term, classification loss, and cycle-consistency loss.

Application: StarGAN



The figure is from Choi et al. (2018).

Lecture Topics

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Likelihood Maximization Principle
4. Basics of Probabilistic Modelling

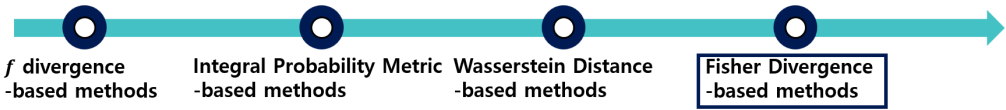
M2. Likelihood Ratio-based Generative Models

5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

M3. Generative Models with Other Statistical Distance

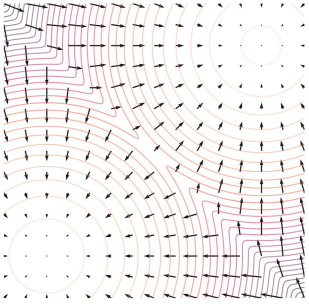
9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Fisher Divergence



- IPM and Wasserstein distance-based methods have alleviated optimization issues. However, adversarial training is still practically difficult.
- Recent works have focused on score functions instead of densities, using estimated scores to generate data.
- Useful materials include Song (2021): `https://yang-song.net/blog/2021/score/`.

Fisher Divergence

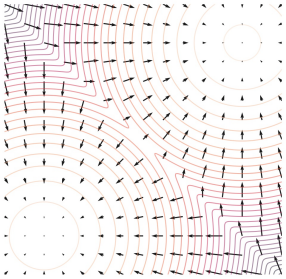


- Fisher divergence (Johnson, 2004) is the expected difference between the (Stein) scores (Liu et al., 2016) of two distributions. It can be expressed as:

$$\mathrm{FD}(p \parallel p_\theta) = \int_{\mathbb{R}^d} \left\| \nabla_{\boldsymbol{x}} \log p(\boldsymbol{x}) - \nabla_{\boldsymbol{x}} \log p_\theta(\boldsymbol{x}) \right\|^2 p(\boldsymbol{x})\, d\boldsymbol{x}. \tag{49}$$

Visualization of $\nabla_{\boldsymbol{x}} \log p(\boldsymbol{x})$ of a mixture of two Gaussians, from Song (2021).

Fisher Divergence



- It is zero if and only if $p = q$.
  Proof:

$$\nabla_{\boldsymbol{x}} \log p(\boldsymbol{x}) = \nabla_{\boldsymbol{x}} \log q(\boldsymbol{x}) \implies p(\boldsymbol{x}) = Cq(\boldsymbol{x}), \tag{50}$$

and $C = 1$ because $\int_{\mathbb{R}^d} p(\boldsymbol{x})\, d\boldsymbol{x} = \int_{\mathbb{R}^d} q(\boldsymbol{x})\, d\boldsymbol{x} = 1$.

Visualization of $\nabla_{\boldsymbol{x}} \log p(\boldsymbol{x})$ of a mixture of two Gaussians, from Song (2021).

Fisher Divergence

- The Fisher divergence is useful for learning energy-based models (Teh et al., 2003), such as Boltzmann distributions:

$$p_\theta(\boldsymbol{x}) = \frac{1}{C(\theta)} \exp(-E_\theta(\boldsymbol{x})), \tag{51}$$

where $C(\theta) := \int_{\mathbb{R}^d} \exp(-E_\theta(\boldsymbol{x}))\, d\boldsymbol{x}$. In this case,

$$\nabla_{\boldsymbol{x}} \log p_\theta(\boldsymbol{x}) = -\nabla_{\boldsymbol{x}} E_\theta(\boldsymbol{x}) \tag{52}$$

holds, and the normalizing constant $C(\theta)$ disappears.

Fisher Divergence

Exercise 10

Let $p_{\theta_0}(\boldsymbol{x})$ and $p_\theta(\boldsymbol{x})$ be PDFs of $\mathcal{N}(\boldsymbol{\mu}_0, \sigma^2 I_d)$ and $\mathcal{N}(\boldsymbol{\mu}, \sigma^2 I_d)$, respectively.

1. Show that the Stein score of $p_\theta(\boldsymbol{x})$ is $\nabla_{\boldsymbol{x}} \log p_\theta(\boldsymbol{x}) = -\frac{1}{\sigma^2}(\boldsymbol{x} - \boldsymbol{\mu})$.

Fisher Divergence

2. Show that $\mathrm{FD}(p_{\theta_0} \parallel p_\theta) = \|\boldsymbol{\mu}_0 - \boldsymbol{\mu}\|_2^2/(\sigma^4)$.

Fisher Divergence and Kullback-Leibler Divergence

- Note that $\nabla_{\boldsymbol{x}} \log p_\theta(\boldsymbol{x})$ differs from $\nabla_\theta \log p_\theta(\boldsymbol{x})$ $(= \partial \log p_\theta(\boldsymbol{x})/\partial\theta)$ in maximum likelihood estimation.
- While MLEs minimize the KL-divergence between model and true distributions, score matching estimators minimize the change of the KL-divergence w.r.t. a white noise:

$$\mathrm{FD}(p(\boldsymbol{x}) \parallel p_\theta(\boldsymbol{x})) = \lim_{t \to 0^+} \frac{\mathcal{D}_{\mathrm{KL}}\big(p(\boldsymbol{x} + \sqrt{t}\boldsymbol{w}) \parallel p_\theta(\boldsymbol{x} + \sqrt{t}\boldsymbol{w})\big) - \mathcal{D}_{\mathrm{KL}}\big(p(\boldsymbol{x}) \parallel p_\theta(\boldsymbol{x})\big)}{t} \tag{53}$$

  where $\boldsymbol{w}$ represents a white Gaussian noise (up to a sign-preserving constant).
- See Theorem 1 in Lyu (2009) for details. Under certain conditions, the minimizer of the Fisher-divergence is consistent (Hyvärinen, 2005).

Data Generation using Score Function

- When we model $p_\theta(\boldsymbol{x})$ as a parametric family distribution and use the minimizer of $\mathrm{FD}(p \parallel p_\theta)$ as an estimator, we can directly generate data from $p_\theta(\boldsymbol{x})$ with the estimator.
- How to generate data using score functions $S(\boldsymbol{x})$ in general?
- The key idea is to introduce *Langevin dynamics* in the sampling process.

Data Generation using Score Function

- Langevin dynamics describes the stochastic movement of a fluid particle located at $\boldsymbol{X}(t)$:

$$m\frac{d^2\boldsymbol{X}(t)}{dt^2} = -\nabla_{\boldsymbol{x}=\boldsymbol{X}(t)}U(\boldsymbol{x}) - \lambda\frac{d\boldsymbol{X}(t)}{dt} + \sqrt{2\lambda k_B T}\boldsymbol{B}(t), \tag{54}$$

  where m is the mass, U is the potential function, λ is the damping coefficient, k_B is the Boltzmann constant, T is the temperature, and $\boldsymbol{B}(t)$ represents the Brownian motion.

- In the overdamped case, where the inertial force is negligible, when $\lambda = 1$, we get:

$$d\boldsymbol{X}(t) = -\nabla_{\boldsymbol{x}=\boldsymbol{X}(t)}U(\boldsymbol{x})\,dt + \sqrt{2k_B T}\,d\boldsymbol{B}(t), \tag{55}$$

  where $d\boldsymbol{B}(t) \sim \mathcal{N}(0, dt\boldsymbol{I}_d)$.[7] Its stationary distribution is the Boltzmann distribution $p(\boldsymbol{x}(\infty)) \propto \exp\big(-U(\boldsymbol{x}(\infty))/(k_B T)\big)$.

[7] This is a special case of the Itô drift-diffusion process.

Data Generation using Score Function

- By substituting $U(\boldsymbol{x}) = -\log p(\boldsymbol{x})$ and setting $T = 1/k_B$, we obtain:

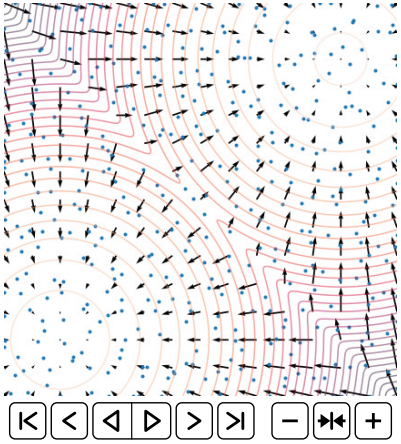
$$d\boldsymbol{X}(t) = \nabla_{\boldsymbol{x}} \log p(\boldsymbol{x})\, dt + \sqrt{2dt}\, \boldsymbol{\mathcal{E}}(t), \quad \boldsymbol{\mathcal{E}}(t) \sim \mathcal{N}(0, \boldsymbol{I}_d), \tag{56}$$

  and the corresponding stationary distribution is $p(\boldsymbol{x})$.
- The discrete approximation with $dt = \eta/2$ and $S_{\theta^*}(\boldsymbol{x})$ results in the following iterative sampling process:

$$\boldsymbol{X}(t) = \boldsymbol{X}(t-1) + \frac{\eta}{2} S_{\theta^*}(\boldsymbol{X}(t-1)) + \sqrt{\eta}\, \boldsymbol{\mathcal{E}}(t), \tag{57}$$

  where $\boldsymbol{X}(T)$ approximately follows $p(\boldsymbol{x})$.
- See: (1) Dr. Kirill Neklyudov's lecture (`https://www.youtube.com/watch?v=3-KzIjoFJy4`) and (2) "Recovering Boltzmann Statistics" (`https://en.wikipedia.org/wiki/Langevin_equation`).

Data Generation using Score Function



The video is from Song (2021).

Score Matching Estimation

- Score matching estimation (Hyvärinen, 2005) was proposed targeting Fisher divergence in learning distributions.
- Let $S_\theta(\boldsymbol{x}) := \nabla_{\boldsymbol{x}} \log p_\theta(\boldsymbol{x})$. Then,

$$\mathrm{FD}(p \parallel p_\theta) = \int_{\mathbb{R}^d} \left(\mathrm{tr}(\nabla_{\boldsymbol{x}} S_\theta(\boldsymbol{x})) + \frac{1}{2} \| S_\theta(\boldsymbol{x}) \|^2 \right) p(\boldsymbol{x}) \, d\boldsymbol{x} \tag{58}$$

up to a constant addition and sign-preserving multiplication. We assume that $S_\theta(\boldsymbol{x}) p(\boldsymbol{x})$ vanishes at the boundary, e.g., $(x_1, \ldots, x_{i-1}, \pm\infty, x_{i+1}, \ldots, x_d)$.

Score Matching Estimation

- **Proof**:

$$\begin{aligned}\mathrm{FD}(p \parallel p_\theta) &:= \int_{\mathbb{R}^d} \|\nabla_{\boldsymbol{x}} \log p(\boldsymbol{x}) - S_\theta(\boldsymbol{x})\|^2 p(\boldsymbol{x})\, d\boldsymbol{x} \\ &= C - 2\int_{\mathbb{R}^d} \big(S_\theta(\boldsymbol{x})^\top \nabla_{\boldsymbol{x}} \log p(\boldsymbol{x})\big) p(\boldsymbol{x})\, d\boldsymbol{x} + \int_{\mathbb{R}^d} \|S_\theta(\boldsymbol{x})\|^2 p(\boldsymbol{x})\, d\boldsymbol{x}.\end{aligned} \tag{59}$$

- Here, $\int_{\mathbb{R}^d} \big(S_\theta(\boldsymbol{x})^\top \nabla_{\boldsymbol{x}} \log p(\boldsymbol{x})\big) p(\boldsymbol{x})\, d\boldsymbol{x}$ equals $-\int_{\mathbb{R}^d} \mathrm{tr}(\nabla_{\boldsymbol{x}} S_\theta(\boldsymbol{x})) p(\boldsymbol{x})\, d\boldsymbol{x}$.

Score Matching Estimation

- **Proof (Cont.)**: Let $\boldsymbol{x}_{-i} := (x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_d)^\top$. Then,

$$\begin{aligned}\int_{\mathbb{R}^d} \left(S_\theta(\boldsymbol{x})^\top \nabla_{\boldsymbol{x}} \log p(\boldsymbol{x})\right) p(\boldsymbol{x})\, d\boldsymbol{x} &= \int_{\mathbb{R}^d} S_\theta(\boldsymbol{x})^\top \nabla_{\boldsymbol{x}} p(\boldsymbol{x})\, d\boldsymbol{x} \\ &= \sum_{i=1}^{d} \int_{\mathbb{R}^{d-1}} \left(\int_{-\infty}^{\infty} S_\theta(\boldsymbol{x})_i \frac{\partial}{\partial x_i} p(\boldsymbol{x})\, dx_i \right) d\boldsymbol{x}_{-i}. \end{aligned} \tag{60}$$

- Since $S_\theta(\boldsymbol{x})p(\boldsymbol{x})$ vanishes at the boundary, by partial integration, we have:

$$\int_{-\infty}^{\infty} S_\theta(\boldsymbol{x})_i \frac{\partial}{\partial x_i} p(\boldsymbol{x})\, dx_i = -\int_{-\infty}^{\infty} \left(\frac{\partial}{\partial x_i} S_\theta(\boldsymbol{x})_i \right) p(\boldsymbol{x})\, dx_i. \tag{61}$$

Thus, $\mathrm{FD}(p \parallel p_\theta) = C + \int_{\mathbb{R}^d} \left(2 \operatorname{tr}(\nabla_{\boldsymbol{x}} S_\theta(\boldsymbol{x})) + \|S_\theta(\boldsymbol{x})\|^2\right) p(\boldsymbol{x})\, d\boldsymbol{x}$, which concludes the proof.

Score Matching Estimation

Exercise 11

Consider the probability density function of $\mathcal{N}(\boldsymbol{\mu}, \sigma^2 I_d)$, $p_\theta(\boldsymbol{x})$ with unknown mean $\theta = \boldsymbol{\mu}$ and known variance σ^2.

1. Show that $\text{tr}(\nabla_{\boldsymbol{x}} S_\theta(\boldsymbol{x})) = -d/\sigma^2$.

Score Matching Estimation

2. Given observations $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^d$, find the score matching estimator $\hat{\boldsymbol{\mu}}_n^{\mathrm{SM}}$ that minimizes the empirical score matching objective:

$$\hat{J}(\boldsymbol{\mu}) := \frac{1}{n} \sum_{i=1}^{n} \left(\mathrm{tr}(\nabla_{\boldsymbol{x}} S_\theta(\boldsymbol{x}_i)) + \frac{1}{2} \left\| S_\theta(\boldsymbol{x}_i) \right\|_2^2 \right). \tag{62}$$

Sliced Score Matching

- In the objective of score matching estimation, $\int \left(\mathrm{tr}(\nabla_{\boldsymbol{x}} S_\theta(\boldsymbol{x})) + \frac{1}{2} \|S_\theta(\boldsymbol{x})\|^2 \right) p(\boldsymbol{x})\, d\boldsymbol{x}$, the Hessian term $\nabla_{\boldsymbol{x}} S_\theta(\boldsymbol{x})$ poses another computational challenge.
- Sliced score matching (Song et al., 2020) targets sliced Fisher divergence (SFD),

$$\mathrm{SFD}(p \parallel p_\theta) := \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \left(\boldsymbol{v}^\top \nabla_{\boldsymbol{x}} \log p(\boldsymbol{x}) - \boldsymbol{v}^\top \nabla_{\boldsymbol{x}} \log p_\theta(\boldsymbol{x}) \right)^2 p(\boldsymbol{x}) p(\boldsymbol{v})\, d\boldsymbol{x}\, d\boldsymbol{v}, \tag{63}$$

  to overcome this limitation.
- The SFD is the average difference between randomly projected scores.

Sliced Score Matching

- In a similar way used in score matching estimation,

$$\mathrm{SFD}(p \parallel p_\theta) = \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \left(\boldsymbol{v}^\top \nabla_{\boldsymbol{x}} S_\theta(\boldsymbol{x}) \boldsymbol{v} + \frac{1}{2} \left(\boldsymbol{v}^\top S_\theta(\boldsymbol{x}) \right)^2 \right) p(\boldsymbol{x}) p(\boldsymbol{v}) \, d\boldsymbol{x} \, d\boldsymbol{v} \tag{64}$$

  up to a constant addition and sign-preserving multiplication.
- By changing the target statistical distances from FD to SFD, the computational bottleneck shifts from computing $\mathrm{tr}(\nabla_{\boldsymbol{x}} S_\theta(\boldsymbol{x}))$ to computing $\boldsymbol{v}^\top \nabla_{\boldsymbol{x}} S_\theta(\boldsymbol{x}) = \nabla_{\boldsymbol{x}} \left(\boldsymbol{v}^\top S_\theta(\boldsymbol{x}) \right)^\top$, which is numerically less demanding.

Sliced Score Matching

- When $p(\boldsymbol{v})$ is the multivariate standard Gaussian distribution, the equation

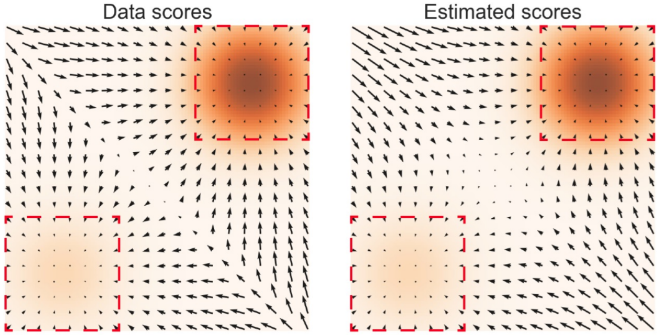
$$\int_{\mathbb{R}^d} \left(\boldsymbol{v}^\top S_\theta(\boldsymbol{x})\right)^2 p(\boldsymbol{v})\, d\boldsymbol{v} = \|S_\theta(\boldsymbol{x})\|^2 \tag{65}$$

  holds, further reducing the computational cost.
- Furthermore, under certain conditions, the minimizer of SFD is consistent and holds asymptotic normality (Song et al., 2020).

Noise Conditional Score Network

- Noise Conditional Score Networks (NCSNs) (Song and Ermon, 2019) are score-based generative models that use estimated scores $S_\theta(\boldsymbol{x})$ to generate data.
- NCSNs introduce a U-Net-based network to model $S(\boldsymbol{x})$. Note that the shape of $S(\boldsymbol{x}) := \nabla_{\boldsymbol{x}} \log p(\boldsymbol{x})$ is the same as that of $\boldsymbol{x}$.

Noise Conditional Score Network



- The initial points are likely to lie in low-density regions, and estimation performances on those regions may be poor.
  Q: The existence of low-density regions is not a new phenomenon compared to other generative models. Why can it be critical in score-based approaches?

The figure is from Song and Ermon (2019). Two red dashed-rectangles display regions where $S_\theta(\boldsymbol{x}) \approx S(\boldsymbol{x})$.

Noise Conditional Score Network: Training

- NCSNs employ the denoising score matching method (Vincent, 2011). They add noise to the data, $\boldsymbol{X} + \sigma\boldsymbol{\mathcal{E}}$, learn its score $S_\theta(\boldsymbol{x}; \sigma)$, and use $S_\theta(\boldsymbol{x}; \sigma)$ with a sufficiently small σ for effective sampling.
- For a given noise level σ, the objective can be expressed as:

$$l(\theta; \sigma) := \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \left\| S_\theta(\boldsymbol{x} + \sigma\boldsymbol{\epsilon}, \sigma) - \nabla_{\boldsymbol{x}+\sigma\boldsymbol{\epsilon}} \log p(\boldsymbol{x} + \sigma\boldsymbol{\epsilon} \mid \boldsymbol{x}) \right\|^2 \times \mathcal{N}(\boldsymbol{\epsilon}; 0, \boldsymbol{I}_d) p(\boldsymbol{x})\, d\boldsymbol{\epsilon}\, d\boldsymbol{x}. \tag{66}$$

Noise Conditional Score Network: Training

Exercise 12

For a given datum $\boldsymbol{x} \in \mathbb{R}^d$, noise vector $\boldsymbol{\epsilon} \in \mathbb{R}^d$, and noise level $\sigma > 0$, let $\tilde{\boldsymbol{x}} = \boldsymbol{x} + \sigma\boldsymbol{\epsilon}$, so that the $p(\tilde{\boldsymbol{x}} \mid \boldsymbol{x})$ is $\mathcal{N}(\tilde{\boldsymbol{x}}; \boldsymbol{x}, \sigma^2 I_d)$. Show that the denoising score matching loss can be expressed as:

$$\|S_\theta(\boldsymbol{x} + \sigma\boldsymbol{\epsilon}, \sigma) + \boldsymbol{\epsilon}/\sigma\|_2^2. \tag{67}$$

Noise Conditional Score Network: Training

Noise Conditional Score Network: Training

- To train the score network across multiple noise scales simultaneously, NCSNs specify a geometric sequence of noise levels $(\sigma_l := \sigma_1 \gamma^{l-1})_{l=1}^{L}$.
- Combining the denoising score matching losses $l(\theta; \sigma_l)$ across all noise levels yields the final objective:

$$\sum_{l=1}^{L} \lambda_l \, l(\theta; \sigma_l), \tag{68}$$

where $(\lambda_l)_{l=1}^{L}$ represents weighting coefficients (e.g., $\lambda_l = \sigma_l^2$) corresponding to the noise levels $(\sigma_l)_{l=1}^{L}$.

Noise Conditional Score Network: Generation

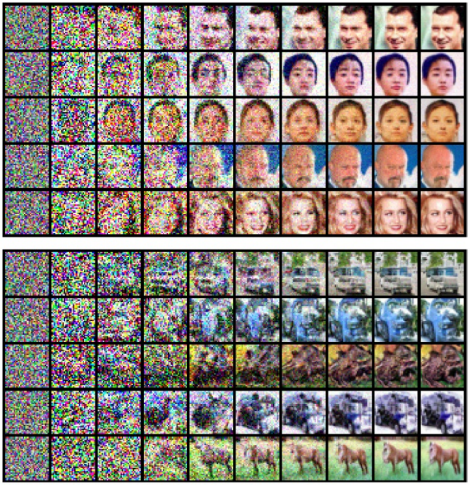
Algorithm 1 Annealed Langevin dynamics.

Require: $\{\sigma_i\}_{i=1}^{L}, \epsilon, T$.

Initialize $\tilde{\mathbf{x}}_0$
for $i \leftarrow 1$ to L **do**
  $\alpha_i \leftarrow \epsilon \cdot \sigma_i^2/\sigma_L^2$ ▷ α_i is the step size.
  for $t \leftarrow 1$ to T **do**
    Draw $\mathbf{z}_t \sim \mathcal{N}(0, I)$
    $\tilde{\mathbf{x}}_t \leftarrow \tilde{\mathbf{x}}_{t-1} + \frac{\alpha_i}{2}\mathbf{s}_{\boldsymbol{\theta}}(\tilde{\mathbf{x}}_{t-1}, \sigma_i) + \sqrt{\alpha_i}\,\mathbf{z}_t$
  end for
  $\tilde{\mathbf{x}}_0 \leftarrow \tilde{\mathbf{x}}_T$
end for
return $\tilde{\mathbf{x}}_T$

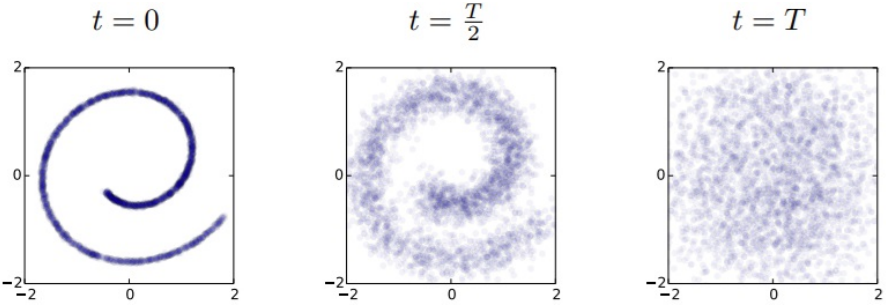
The algorithm is from Song and Ermon (2019).

Noise Conditional Score Network: Result



The figure is from Song and Ermon (2019).

Motivation: Diffusion



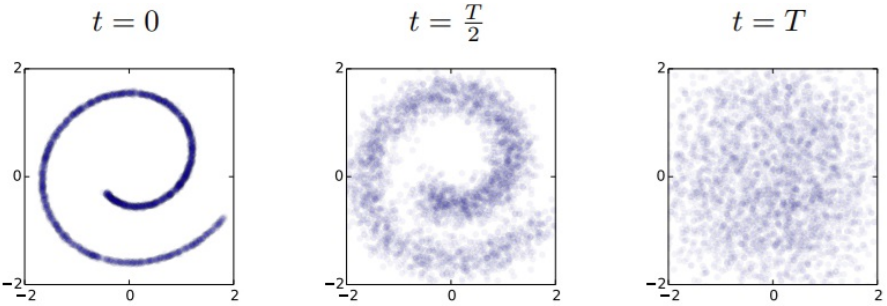
- For a given datum from $p(\boldsymbol{x})$, let's gradually mix it with Gaussian noises:

$$\boldsymbol{x}(t) := \sqrt{1 - \beta_t}\, \boldsymbol{x}(t-1) + \sqrt{\beta_t}\, \boldsymbol{\epsilon}(t), \tag{69}$$

where $\boldsymbol{x}(0)$ represents the initial datum sampled from $p(\boldsymbol{x})$ and $\boldsymbol{\epsilon}(t) \sim \mathcal{N}(0, \boldsymbol{I}_d)$.

The figure is from Sohl-Dickstein et al. (2015).

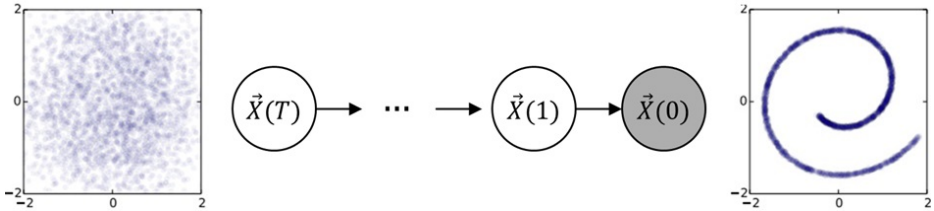
Motivation: Diffusion



- Then, the distribution of $\boldsymbol{x}(T)$ converges to $\mathcal{N}(0, \boldsymbol{I}_d)$ as $T \to \infty$ and $\beta_t \to 0$.
- The key idea of diffusion models is to learn $p(\boldsymbol{x}(0) \mid \boldsymbol{x}(T))$ to generate data.

The figure is from Sohl-Dickstein et al. (2015).

Denoising Diffusion Probabilistic Model: Generation



- The generation process (called "Backward Diffusion Process") can be formulated as:

$$p_\theta(\boldsymbol{x}(0), \ldots, \boldsymbol{x}(T)) = p(\boldsymbol{x}(T)) \prod_{t=1}^{T} p_\theta\big(\boldsymbol{x}(t-1) \mid \boldsymbol{x}(t)\big), \tag{70}$$

where $p(\boldsymbol{x}(T))$ is the density of $\mathcal{N}(0, \boldsymbol{I}_d)$ and $p_\theta(\boldsymbol{x}(t-1) \mid \boldsymbol{x}(t))$ is that of $\mathcal{N}\big(\boldsymbol{\mu}_\theta(\boldsymbol{x}(t), t), \boldsymbol{\Sigma}_\theta(\boldsymbol{x}(t), t)\big)$.

Denoising Diffusion Probabilistic Model: Inference

- Denoising Diffusion Probabilistic Model (DDPM) (Ho et al., 2020) is a generative model based on this idea.
- The data likelihood $p_\theta(\boldsymbol{x}(0))$ is highly intractable. DDPMs apply variational inference.
- The diffusion process gives a closed-form expression of the optimal posterior:[8]

$$q\big(\boldsymbol{x}(t) \mid \boldsymbol{x}(0), \ldots, \boldsymbol{x}(t-1)\big) = q\big(\boldsymbol{x}(t) \mid \boldsymbol{x}(t-1)\big) = \mathcal{N}\big(\boldsymbol{x}(t); \sqrt{1-\beta_t}\boldsymbol{x}(t-1), \beta_t \boldsymbol{I}_d\big). \tag{71}$$

It is called "Forward Diffusion Process". This implies:

$$q\big(\boldsymbol{x}(1), \ldots, \boldsymbol{x}(T) \mid \boldsymbol{x}(0)\big) = \prod_{t=1}^{T} q\big(\boldsymbol{x}(t) \mid \boldsymbol{x}(t-1)\big). \tag{72}$$

[8] In other words, we already know the ground-truth inference process.

Denoising Diffusion Probabilistic Model: Inference

Exercise 13

For variance parameters $0 < \beta_1, \beta_2 < 1$, let $\alpha_t := 1 - \beta_t$ $(t \in \{1, 2\})$ and $\bar{\alpha}_2 := \alpha_1 \alpha_2$. Given a fixed initial vector $\boldsymbol{x}(0) \in \mathbb{R}^d$ and independent standard Gaussian random vectors $\boldsymbol{\mathcal{E}}(1), \boldsymbol{\mathcal{E}}(2) \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \boldsymbol{I}_d)$, consider the two-step forward diffusion process:

$$\begin{aligned} \boldsymbol{X}(1) &= \sqrt{\alpha_1}\, \boldsymbol{x}(0) + \sqrt{1 - \alpha_1}\, \boldsymbol{\mathcal{E}}(1), \\ \boldsymbol{X}(2) &= \sqrt{\alpha_2}\, \boldsymbol{X}(1) + \sqrt{1 - \alpha_2}\, \boldsymbol{\mathcal{E}}(2). \end{aligned} \tag{73}$$

Show that $\boldsymbol{X}(2) \sim \mathcal{N}\big(\sqrt{\bar{\alpha}_2}\, \boldsymbol{x}(0), (1 - \bar{\alpha}_2) \boldsymbol{I}_d\big)$.

Denoising Diffusion Probabilistic Model: Inference



Denoising Diffusion Probabilistic Model: Loss

- The loss function is the negative ELBO w.r.t. $\log p_\theta(\boldsymbol{x}(0))$:

$$-\log p_\theta(\boldsymbol{x}(0)) + \mathcal{D}_{\mathrm{KL}}\Big(q\big(\boldsymbol{x}(1),\ldots,\boldsymbol{x}(T) \mid \boldsymbol{x}(0)\big)\Big\|p_\theta\big(\boldsymbol{x}(1),\ldots,\boldsymbol{x}(T) \mid \boldsymbol{x}(0)\big)\Big). \tag{74}$$

- This negative ELBO can be expressed as:

$$-\int_{\mathbb{R}^d}\big(\log p_\theta(\boldsymbol{x}(0) \mid \boldsymbol{x}(1))\big)q(\boldsymbol{x}(1) \mid \boldsymbol{x}(0))\,d\boldsymbol{x}(1) + \mathcal{D}_{\mathrm{KL}}\big(q(\boldsymbol{x}(T) \mid \boldsymbol{x}(0)) \parallel p(\boldsymbol{x}(T))\big)$$
$$+\sum_{t=2}^{T}\int_{\mathbb{R}^d}\mathcal{D}_{\mathrm{KL}}\Big(q\big(\boldsymbol{x}(t-1) \mid \boldsymbol{x}(0),\boldsymbol{x}(t)\big)\Big\|p_\theta\big(\boldsymbol{x}(t-1) \mid \boldsymbol{x}(t)\big)\Big)q(\boldsymbol{x}(t) \mid \boldsymbol{x}(0))\,d\boldsymbol{x}(t). \tag{75}$$

Denoising Diffusion Probabilistic Model: Loss

Proof:

- The ELBO can be written as:

$$
\begin{aligned}
&\int \big(\log p_\theta(\boldsymbol{x}(0) \mid \boldsymbol{x}(1), \ldots, \boldsymbol{x}(T))\big) q(\boldsymbol{x}(1), \ldots, \boldsymbol{x}(T) \mid \boldsymbol{x}(0))\, d\boldsymbol{x}(1) \cdots d\boldsymbol{x}(T) \\
&\quad - \mathcal{D}_{\mathrm{KL}}\Big(q(\boldsymbol{x}(1), \ldots, \boldsymbol{x}(T) \mid \boldsymbol{x}(0)) \Big\| p_\theta(\boldsymbol{x}(1), \ldots, \boldsymbol{x}(T))\Big) \\
&= \int \big(\log p_\theta(\boldsymbol{x}(0) \mid \boldsymbol{x}(1))\big) q(\boldsymbol{x}(1) \mid \boldsymbol{x}(0))\, d\boldsymbol{x}(1) \\
&\quad - \mathcal{D}_{\mathrm{KL}}\Big(q(\boldsymbol{x}(1), \ldots, \boldsymbol{x}(T) \mid \boldsymbol{x}(0)) \Big\| p_\theta(\boldsymbol{x}(1), \ldots, \boldsymbol{x}(T))\Big).
\end{aligned}
\tag{76}
$$

- In the last KL term, the second argument can be expressed as $p(\boldsymbol{x}(T)) \prod_{t=2}^{T} p_\theta(\boldsymbol{x}(t-1) \mid \boldsymbol{x}(t))$.

Denoising Diffusion Probabilistic Model: Loss

Proof (Cont.):

- In the last KL term, the first argument can be expressed as $\prod_{t=1}^{T} q(\boldsymbol{x}(t) \mid \boldsymbol{x}(t-1))$. Here, by Bayes' Theorem:

$$q\big(\boldsymbol{x}(t-1) \mid \boldsymbol{x}(0), \boldsymbol{x}(t)\big) = \frac{q(\boldsymbol{x}(t) \mid \boldsymbol{x}(0), \boldsymbol{x}(t-1)) q(\boldsymbol{x}(t-1) \mid \boldsymbol{x}(0))}{q(\boldsymbol{x}(t) \mid \boldsymbol{x}(0))}. \tag{77}$$

- Note that the Forward Diffusion Process structure implies $q(\boldsymbol{x}(t) \mid \boldsymbol{x}(0), \boldsymbol{x}(t-1)) = q(\boldsymbol{x}(t) \mid \boldsymbol{x}(t-1))$, yielding for $t > 1$:

$$q\big(\boldsymbol{x}(t) \mid \boldsymbol{x}(t-1)\big) = q\big(\boldsymbol{x}(t-1) \mid \boldsymbol{x}(0), \boldsymbol{x}(t)\big) \frac{q(\boldsymbol{x}(t) \mid \boldsymbol{x}(0))}{q(\boldsymbol{x}(t-1) \mid \boldsymbol{x}(0))}. \tag{78}$$

Denoising Diffusion Probabilistic Model: Loss

Proof (Cont.):

- These imply:

$$
\begin{aligned}
&\log \frac{q(\boldsymbol{x}(1), \ldots, \boldsymbol{x}(T) \mid \boldsymbol{x}(0))}{p_\theta(\boldsymbol{x}(1), \ldots, \boldsymbol{x}(T))} \\
&= \log \frac{\prod_{t=1}^{T} q(\boldsymbol{x}(t) \mid \boldsymbol{x}(t-1))}{p(\boldsymbol{x}(T)) \prod_{t=2}^{T} p_\theta(\boldsymbol{x}(t-1) \mid \boldsymbol{x}(t))} \\
&= \log \frac{q(\boldsymbol{x}(1) \mid \boldsymbol{x}(0)) \prod_{t=2}^{T} q(\boldsymbol{x}(t-1) \mid \boldsymbol{x}(0), \boldsymbol{x}(t)) \frac{q(\boldsymbol{x}(t) \mid \boldsymbol{x}(0))}{q(\boldsymbol{x}(t-1) \mid \boldsymbol{x}(0))}}{p(\boldsymbol{x}(T)) \prod_{t=2}^{T} p_\theta(\boldsymbol{x}(t-1) \mid \boldsymbol{x}(t))} \\
&= \log \frac{q(\boldsymbol{x}(T) \mid \boldsymbol{x}(0))}{p(\boldsymbol{x}(T))} + \sum_{t=2}^{T} \log \frac{q(\boldsymbol{x}(t-1) \mid \boldsymbol{x}(0), \boldsymbol{x}(t))}{p_\theta(\boldsymbol{x}(t-1) \mid \boldsymbol{x}(t))}.
\end{aligned}
\tag{79}
$$

Denoising Diffusion Probabilistic Model: Loss

Proof (Cont.):

- Thus,

$$
\begin{aligned}
&\mathcal{D}_{\mathrm{KL}}\Big(q(\boldsymbol{x}(1),\ldots,\boldsymbol{x}(T)\mid\boldsymbol{x}(0))\Big\|p_\theta(\boldsymbol{x}(1),\ldots,\boldsymbol{x}(T))\Big)\\
&=\int\left(\log\frac{q(\boldsymbol{x}(T)\mid\boldsymbol{x}(0))}{p(\boldsymbol{x}(T))}\right)q(\boldsymbol{x}(T)\mid\boldsymbol{x}(0))\,d\boldsymbol{x}(T)\\
&\quad+\sum_{t=2}^{T}\int\left(\log\frac{q(\boldsymbol{x}(t-1)\mid\boldsymbol{x}(0),\boldsymbol{x}(t))}{p_\theta(\boldsymbol{x}(t-1)\mid\boldsymbol{x}(t))}\right)q(\boldsymbol{x}(t-1),\boldsymbol{x}(t)\mid\boldsymbol{x}(0))\,d\boldsymbol{x}(t-1)\,d\boldsymbol{x}(t)\\
&=\mathcal{D}_{\mathrm{KL}}\big(q(\boldsymbol{x}(T)\mid\boldsymbol{x}(0))\parallel p(\boldsymbol{x}(T))\big)\\
&\quad+\sum_{t=2}^{T}\int\mathcal{D}_{\mathrm{KL}}\Big(q(\boldsymbol{x}(t-1)\mid\boldsymbol{x}(0),\boldsymbol{x}(t))\Big\|p_\theta(\boldsymbol{x}(t-1)\mid\boldsymbol{x}(t))\Big)q(\boldsymbol{x}(t)\mid\boldsymbol{x}(0))\,d\boldsymbol{x}(t).
\end{aligned}
\tag{80}
$$

Denoising Diffusion Probabilistic Model: Loss

- Again, the loss function (the negative ELBO) can be expressed as:

$$-\int_{\mathbb{R}^d}\big(\log p_\theta(\boldsymbol{x}(0)\mid\boldsymbol{x}(1))\big)q(\boldsymbol{x}(1)\mid\boldsymbol{x}(0))\,d\boldsymbol{x}(1)+\mathcal{D}_{\mathrm{KL}}\big(q(\boldsymbol{x}(T)\mid\boldsymbol{x}(0))\parallel p(\boldsymbol{x}(T))\big)$$
$$+\sum_{t=2}^{T}\int_{\mathbb{R}^d}\mathcal{D}_{\mathrm{KL}}\Big(q\big(\boldsymbol{x}(t-1)\mid\boldsymbol{x}(0),\boldsymbol{x}(t)\big)\Big\|p_\theta\big(\boldsymbol{x}(t-1)\mid\boldsymbol{x}(t)\big)\Big)q(\boldsymbol{x}(t)\mid\boldsymbol{x}(0))\,d\boldsymbol{x}(t).$$

- Here, $q(\boldsymbol{x}(t-1)\mid\boldsymbol{x}(0),\boldsymbol{x}(t))$ is the density of $\mathcal{N}\big(\boldsymbol{\mu}_t(\boldsymbol{x}(0),\boldsymbol{x}(t)),\bar{\beta}_t\boldsymbol{I}_d\big)$ (Ho et al., 2020), where $\bar{\alpha}_t:=\prod_{s=1}^{t}(1-\beta_s)$, $\bar{\beta}_t:=\frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$, and:

$$\boldsymbol{\mu}_t(\boldsymbol{x}(0),\boldsymbol{x}(t))=\frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\boldsymbol{x}(0)+\frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\boldsymbol{x}(t). \tag{81}$$

Denoising Diffusion Probabilistic Model: Loss

- When $p_\theta(\boldsymbol{x}(t-1) \mid \boldsymbol{x}(t))$ is the density of $\mathcal{N}\big(\boldsymbol{\mu}_\theta(\boldsymbol{x}(t), t), \bar{\beta}_t \boldsymbol{I}_d\big)$, the KL term can be expressed (up to a constant addition) as:

$$\mathcal{D}_{\mathsf{KL}}\Big(q\big(\boldsymbol{x}(t-1) \mid \boldsymbol{x}(0), \boldsymbol{x}(t)\big)\Big\|p_\theta\big(\boldsymbol{x}(t-1) \mid \boldsymbol{x}(t)\big)\Big) = \frac{1}{2\bar{\beta}_t}\big\|\boldsymbol{\mu}_t(\boldsymbol{x}(0), \boldsymbol{x}(t)) - \boldsymbol{\mu}_\theta(\boldsymbol{x}(t), t)\big\|^2. \tag{82}$$

- Note that $\boldsymbol{x}(t) = \sqrt{\bar{\alpha}_t}\boldsymbol{x}(0) + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}$ for some Gaussian noise $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \boldsymbol{I}_d)$, implying that:

$$\boldsymbol{\mu}_t(\boldsymbol{x}(0), \boldsymbol{x}(t)) = \frac{1}{\sqrt{\alpha_t}}\left(\boldsymbol{x}(t) - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}\right). \tag{83}$$

  Thus, learning $\boldsymbol{\mu}_\theta(\boldsymbol{x}(t), t)$ can be re-formulated as estimating $\boldsymbol{\epsilon}$ using $(\boldsymbol{x}(t), t)$.

Denoising Diffusion Probabilistic Model: Loss

- Motivated by this, DDPMs introduce a noise prediction network $\boldsymbol{\epsilon}_\theta(\boldsymbol{x}(t), t) = \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\boldsymbol{x}(0) + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t)$. See Section 3.2 in Ho et al. (2020) for further details.
- Though DDPMs utilize variational inference, placing them in the category of likelihood-based approaches, they are equivalent to score-based generative models.
- The estimated $\boldsymbol{\epsilon}_\theta(\boldsymbol{x}(t), t)$ can be interpreted as an estimate of score functions of data with Gaussian noises, so $\boldsymbol{\epsilon}_\theta$ can be used to sample data via Langevin dynamics.

Denoising Diffusion Probabilistic Model: Algorithm

Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \text{Uniform}(\{1, \dots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on

$$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$$

6: **until** converged

Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \dots, 1$ **do**
3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

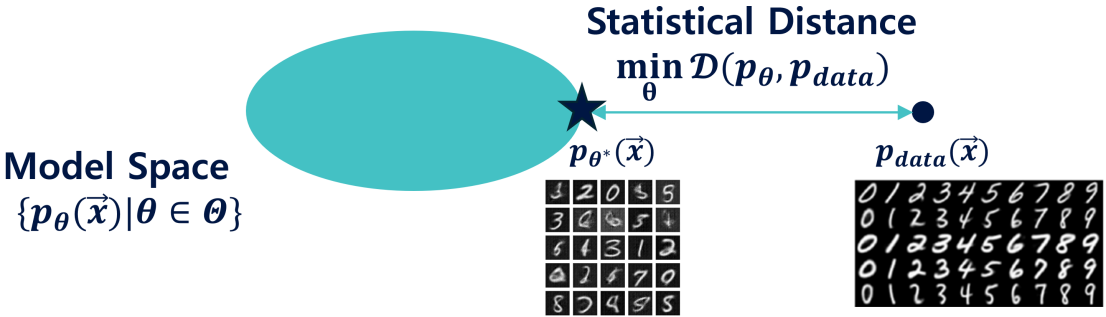
Algorithm tables are from Ho et al. (2020).

Denoising Diffusion Probabilistic Model: Result

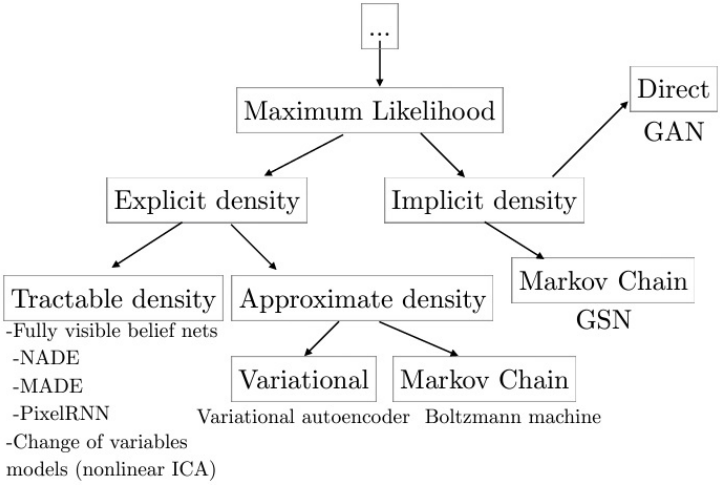


Images are from Ho et al. (2020).

Concluding Remark

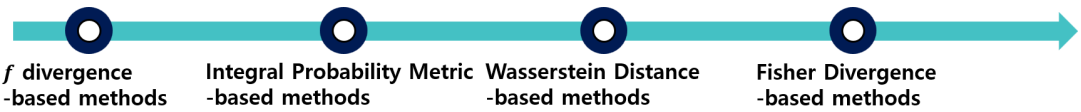


Concluding Remark



The figure is from Goodfellow (2016).

Concluding Remark



- We have reviewed recent developments in deep generative models, with a particular focus on targeted statistical distances.
- There are many ways to explore recent advancements in deep generative models. Examples include developing model classes for specific data types, analyzing the properties of model classes and estimators, or adapting large pre-trained models to new datasets.

Concluding Remark

- Such changes may not always be exciting for everyone.
- Rapid advancements inevitably bring anxiety and uncertainty as the expectations of academia and industry continue to evolve.[9]
- I hope that this course has helped you find joy and beauty in deep generative models and has also encouraged you to continue exploring new waves in this evolving field.

[9] *"i sensed anxiety and frustration at NeurIPS'24"*
https://kyunghyuncho.me/i-sensed-anxiety-and-frustration-at-neurips24/

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